



## Review

## Clogging in subsurface-flow treatment wetlands: Occurrence and contributing factors

Paul Knowles<sup>a</sup>, Gabriela Dotro<sup>b,\*</sup>, Jaime Nivala<sup>c,d</sup>, Joan García<sup>e</sup><sup>a</sup> Sustainable Environment Research Group, Aston University, Birmingham B4 7ET, United Kingdom<sup>b</sup> School of Applied Sciences, Cranfield University, Cranfield, Bedfordshire MK43 0AL, United Kingdom<sup>c</sup> Aarhus University, Department of Biological Sciences, Ole Worms Allé, Building 1135, 8000 Aarhus C, Denmark<sup>d</sup> Helmholtz Center for Environmental Research (UFZ), Environmental and Biotechnology Center (UBZ), Permoserstraße 15, 04318 Leipzig, Germany<sup>e</sup> Environmental Engineering Division, Hydraulics, Coastal and Environmental Engineering Department, Technical University of Catalonia c/Jordi Girona 1-3, Mòdul D-1, 08034 Barcelona, Spain

## ARTICLE INFO

## Article history:

Received 8 October 2009

Received in revised form 13 August 2010

Accepted 23 August 2010

Available online 8 October 2010

## Keywords:

Clog matter

Constructed wetland

Design

Horizontal flow

Vertical flow

Loading rates

Review

## ABSTRACT

Clogging is a major operational and maintenance issue associated with the use of subsurface flow wetlands for wastewater treatment, and can ultimately limit the lifetime of the system. This review considers over two decades of accumulated knowledge regarding clogging in both vertical and horizontal subsurface flow treatment wetlands. The various physical, chemical and biological factors responsible for clogging are identified and discussed. The occurrence of clogging is placed into the context of various design and operational parameters such as wastewater characteristics, upstream treatment processes, intermittent or continuous operation, influent distribution, and media type. This information is then used to describe how clogging develops within, and subsequently impacts, common variants of subsurface flow treatment wetland typically used in the U.S., U.K., France and Germany. Comparison of these systems emphasized that both hydraulic loading rate and solids loading rate need to be considered when designing systems to operate robustly, i.e. hydraulic overloading makes horizontal-flow tertiary treatment systems in the U.K. more susceptible to clogging problems than vertical-flow primary treatment systems in France. Future research should focus on elucidating the underlying mechanisms of clogging as they relate to the design, operation, and maintenance of subsurface flow treatment wetlands.

© 2010 Elsevier B.V. All rights reserved.

## 1. Introduction

Subsurface flow treatment wetlands are used worldwide for removing pollutants from wastewaters due to their mechanical simplicity and low operations and maintenance (O&M) requirements in comparison to conventional wastewater treatment technologies. Subsurface Flow (SSF) Treatment Wetlands (TWs) consist of a bed of porous media planted with emergent macrophytes, through which wastewater is passed for quality improvement. The flow through SSF TWs can be either horizontal (HSSF) or vertical (VF). Typically, the media in HSSF TWs is permanently saturated, whereas the media in VF TWs goes through cycles of saturation because wastewater is dosed through the system intermittently (Vymazal and Kröpfelová, 2008).

In SSF treatment wetlands, the cumulative biological, chemical, and physical treatment processes may cause gradual clogging of the porous media. Clogging may be accompanied by a decrease in

treatment performance or hydraulic malfunctions such as ponding of wastewater on the surface of the system and bypass of untreated wastewater. These issues will require intervention if satisfactory operation is to be maintained. Advanced clogging may eventually necessitate remediation of the clogged media; thus limiting the asset lifetime of the system (Cooper et al., 2005; Caselles-Osorio and García, 2007; Kadlec and Wallace, 2009; Nivala and Rousseau, 2009).

The realization that clogging is a widespread operational problem in SSF TWs has become apparent through frequent reporting over the past two decades (as summarized in this article), in which time thousands of systems have been constructed worldwide. Original predictions of HSSF TW system longevity were on the order of 50–100 years (Conley et al., 1991; Bavor and Schulz, 1993) whereas this estimate has been progressively shortened to 15 years (Cooper et al., 1996), 10 years (Wallace and Knight, 2006) and most recently, to eight years in the U.K. (Griffin et al., 2008). Admittedly, the long-term viability and maintenance requirements of these systems are still unknown (Kadlec and Wallace, 2009). As such, there is motivation to understand and control clogging. The desire to mitigate clogging has directly influenced the design evolution of SSF TWs

\* Corresponding author. Tel.: +44 758 864 5822.

E-mail address: [g.c.dotro@cranfield.ac.uk](mailto:g.c.dotro@cranfield.ac.uk) (G. Dotro).

**Table 1**  
Non-hydrous components of clog matter categorized into whether the accumulation is an intentional part of the wastewater treatment process, or an incidental result of the wastewater treatment process. Incidental accumulations include accidental operations, which are italicized.

| Component        | Intentional accumulation (external loads)                                                              | Incidental accumulation (internal loads)                                                                                                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organic solids   | <ul style="list-style-type: none"> <li>• Wastewater solids</li> </ul>                                  | <ul style="list-style-type: none"> <li>• Biomass growth</li> <li>• Plant roots</li> <li>• Biofilm and plant detritus</li> <li>• <i>Solids introduced during construction</i></li> <li>• Solids from chemical erosion of gravel</li> <li>• <i>Solids introduced during construction</i></li> </ul> |
| Inorganic solids | <ul style="list-style-type: none"> <li>• Wastewater solids</li> <li>• Chemical precipitates</li> </ul> |                                                                                                                                                                                                                                                                                                   |

(Brix and Schierup, 1989; Murphy and Cooper, 2010); however, the factors that lead to clogging are still not well understood. This problem precludes wetland design documents from addressing prevention or long-term management strategies.

This review summarizes the current knowledge and experience on the factors that cause clogging (Section 2), and the influence of design and operational parameters (Section 3). Section 4 explores how clogging influences SSF TW hydraulics and conveys some of the guidelines that are recommended for clogging mitigation. Based on the reviewed content, Section 5 differentiates how clogging manifests in different SSF TW design variants and offers opinions on why some systems are more prone to clogging than others. This paper does not claim to explain the dynamic mechanism of clogging, as the necessary detailed scientific studies to characterize this are currently lacking in the wetland literature. Rather, it is intended that this document will form a useful reference for wetland researchers, designers and practitioners.

## 2. Factors attributing to clogging

Clogging occurs due to accumulation of materials associated with treatment (e.g., intentional or external loads) and other operational factors (e.g., incidental or internal loads). The quantity and composition of this material, hereafter referred to as *clog matter*, will vary depending on these internal and external loads. Clog matter typically consists of highly hydrated gels and sludge (often more than 70%, v/v, water) with inorganic and organic solids (IWA, 2000). The typical components of clog matter are categorized in Table 1.

The composition of clog matter is often a lower density than its constituents such that it can effectively reduce pore space in the granular medium (Baveye et al., 1998). It can accumulate both in the subsurface and on the surface of the SSF TW, reducing subsurface hydraulic conductivity in the former case, and preventing infiltration of surface flow through to the subsurface in the latter case. The net accumulation rate of this material is a balance between the external and internal loads (Table 1), and loss due to export and decomposition (Tanner et al., 1998). This determines whether or not the clog matter will create hydraulic issues on the macroscopic level.

This section elucidates the manner in which the factors listed in Table 1 contribute to clogging. The discussion will consider filtration of particulate matter by the porous media, additional solids contributions from microorganisms (biofilms), vegetation contributions and chemical precipitation. The composition and biodegradability of typical clog matter accumulations will also be discussed in this section.

### 2.1. Solids entrapment

Suspended solids are filtered and retained by subsurface flow treatment wetlands via the mechanisms of transport and attachment (Yao et al., 1971). These mechanisms are the same as those responsible for flocculation, whereby transport mechanisms create collisions between particles, and attachment mechanisms cause

these particles to adhere upon impact (Swift and Friedlander, 1964). Wastewater contains solids with a variety of sizes and compositions. A range of different physico-chemical mechanisms are responsible for overall TSS reduction (Table 2).

The balance of the forces listed in Table 2 leads to a range of removal efficiencies for different particle sizes. Kadlec and Wallace (2009) explain that sedimentation of large particles is the major mechanism of solids removal in HSSF TWs, which predominantly takes place in the first few meters of the wetland. It has frequently been observed that the nadir in removal efficiency occurs for particles that are approximately 1–2  $\mu\text{m}$  in diameter, because they are too small to be removed by inertial and other physical effects, but too large to be affected by electrostatic and Brownian forces (Yao et al., 1971; Logan et al., 1995; Zamani and Maini, 2009). Consequently, a large number of wastewater particles fall into this size range (Levine et al., 1991; Tchobanoglous, 1993) and Puigagut et al. (2008) observed that particles in the 0.7–2  $\mu\text{m}$  range were the most abundant in the influent and effluent of a HSSF TW in Spain.

As particles accumulate within the media, the efficacy of subsequent removal is often enhanced due to the reduction in pore space. Particles which are electrostatically attracted to each other are capable of stacking and often form dendrites which protrude into the pore space and increase the likelihood of particle interception (Hubbe et al., 2009).

Particle retention on media surfaces is due to the electrochemical effect of adsorption (summation of electrical double-layer interactions and the dipole interactions known as van der Waals' forces). The strength of attachment depends on the relative charge of particles, media surfaces and bulk fluid (Hermansson, 1999). Detachment of particles can occur for several reasons. A change in the ionic strength of the wastewater may neutralize or overcome the attractive force between particles and surfaces such that they are released back into solution (an effect known as peptization). Hydrodynamic shear forces can lead to sloughing of particles, especially as pores become constricted and interstitial velocities increase as a consequence (Zamani and Maini, 2009). However, this effect is generally confined to closed, pressurized reactors and in gravitational flow systems such as SSF TWs this effect is uncommon, as flow will find an alternative path (such as overland) once faced with hydraulic resistance (Maloszewski et al., 2006).

### 2.2. Biofilm clogging

Transport and attachment of microbes through porous media can be described using the same set of principles outlined for solids (Hermansson, 1999; Tufenkji, 2007). In SSF TWs, most biomass forms as biofilms on the surfaces of the media, with very little matter suspended in the wastewater (Khatiwada and Polprasert, 1999). Further biomass can proliferate once microbes have colonized media surfaces. Several studies conclude that greater biofilm development occurs at the inlet region where the concentration of organic matter in the wastewater is greatest (Ragusa et al., 2004; García et al., 2007; Tietz et al., 2007). Caselles-Osorio and García (2006) found that biofilm clogging in an experimental HSSF TW

**Table 2**

The various physico-chemical mechanisms that are responsible for solids removal in SSF TWs.

| Large particle removal mechanisms |                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sedimentation and buoyancy        | According to Stoke's Law, particles with different density to wastewater will move vertically across the flow-field under the effect of gravity, until they impact a surface                                              |
| Hydrodynamic effects              | Non-uniform hydrodynamics forces across the body of a particle will cause it to drift across the flow-field                                                                                                               |
| Inertial divergence               | Particles with significant inertia may deviate from streamlines as flow diverges around obstacles, and impact a surface                                                                                                   |
| Interception                      | If the streamline conveying a particle is closer than the radius of the particle interception of the media surface will occur                                                                                             |
| Straining and trapping            | Particles that are larger than pore spaces will be strained and particles may also be trapped by media morphological irregularities. Filamentous/fibrous particles are particularly susceptible to these modes of removal |
| Small particle removal mechanisms |                                                                                                                                                                                                                           |
| Brownian motion                   | Colloidal particles are influenced by the thermal forces responsible for Brownian motion which induce random trajectory through the flow field                                                                            |
| Electrostatic forces              | Repulsive or attractive forces between particles in suspension, and particles and media surfaces will influence particle trajectory                                                                                       |
| Bridging                          | Small particles can be removed within relatively large pores if numerous particles arrive simultaneously and block the pore by bridging                                                                                   |
| Coagulation                       | The coagulation of smaller colloids into larger particles promotes their removal through the previously outlined mechanisms                                                                                               |

reduced inlet hydraulic conductivity to 64% of outlet hydraulic conductivity.

The influence of biofilms on hydraulics is highly variable due to their heterogeneous nature (Thullner, 2010). For instance, biofilms often form filamentous colonies or aggregates in saturated conditions (Vandevivere and Baveye, 1992a,b,c; Dupin and McCarty, 1999; Dupin et al., 2001), developing webs across pore spaces that can trap particles more proficiently than uniform biofilm coatings (Mays and Hunt, 2005). Biomass density will vary with accumulation composition and community, which impacts its hydraulic conductivity. The majority of biofilms secrete extracellular polymeric slime (Madigan et al., 2006) that are typically 99% water (Rittman and McCarty, 1980; Tanner and Sukias, 1995). deBeer and Stoodley (1995) expound that these slimes are analogous to gel networks with pore diameters on the nanometric scale, thus making them relatively impermeable (Taylor et al., 1990; Vandevivere and Baveye, 1992c) and proficient at forming associations with other inorganic and organic materials (Baveye et al., 1998; Thullner, 2010). If biofilms on separate gravel particles bridge, then pore plugging will occur and the hydraulic conductivity of the bulk porous media will tend towards the hydraulic conductivity of the biofilm (Wallace and Knight, 2006).

### 2.3. Vegetation contributions

The role of plants in SSF TW clogging is an evolving debate. One of the original assumptions of the Root Zone Method was that root growth would counteract media clogging (Kickuth and Könnemann, 1988), and the tubular structure of the roots would provide a macro-porous network for flow. However, there are very few data in the literature to support this claim. Fisher (1990) observed that root expansion in an Australian field-scale HSSF TW increased the depth of the reactor at the inlet zone by 60 mm over three years, which would serve to counteract clogging. However, early experience with soil-based HSSF TWs in Northern Europe proved that root growth did not preclude hydraulic malfunctions such as overland flow (Netter and Bischofsberger, 1990; Pauly, 1990; Brix, 1994b). According to IWA (2000), a dense macrophyte stand will have 500–5000 g dry weight (DW) per m<sup>2</sup> of subsurface roots and rhizomes, which provides additional surface area for accumulation and occludes one quarter to one third of pore volume in the root-zone. Indeed, the root mat of *Phragmites australis* has been reported to have hydraulic conductivity on the order of 1–20 m/d (Baird et al., 2004), which is approximately two orders of magnitude smaller than the hydraulic conductivity of clean gravels typically used in SSF TWs (US EPA, 2000).

When winter plant die-back occurs, if the accumulation rate of plant detritus exceeds decomposition rates then a net accumulation of plant detritus will occur over time (Kirschner et al., 2001; Asaeda et al., 2002; Rybczyk et al., 2002; Chimney and Pietro, 2006). Root material contributes to subsurface clogging and leaf litter-fall contributes to surface clogging, with Kadlec and Wallace (2009) stating that typically 5–15% of plant detritus is recalcitrant. Decomposition and accumulation rates will vary geographically but, for example, Kadlec and Watson (1993) measured an average surface accumulation of 2410 g dry matter per m<sup>2</sup> leaf litter after three years of operation in a HSSF TW in Kentucky (U.S.). Aboveground, up to half of the standing plant biomass (stems) may be dead, but remain relatively stable due to esters in the plant tissue that resist degradation (Tanner et al., 1998).

While studying dairy wastewater treatment HSSF TWs in New Zealand, Tanner (1994) illustrated the potential for clogging by plant detritus. Organic matter accumulation rates of 1300–3000 g/m<sup>2</sup>·yr were recorded but only 400–1600 g/m<sup>2</sup>·yr could be directly attributed to wastewater loading, and only 4% of pore blockage was due to live root penetration (Tanner, 1994). Moreover, during the same study Tanner and Sukias (1995) compared organic matter accumulation rates in equivalent planted and unplanted beds and found that accumulation rates were 1200–2000 g/m<sup>2</sup>·yr higher in the planted systems. Accumulations of humic substances were two to eight times higher in the top 100 mm of substrate than they were in the lower strata, while generally only 50% greater in the inlet region than at the outlet. Therefore, the authors concluded that the humic contributions from plants, rather than accumulation of wastewater organic matter, were the controlling factor in the clogging of this system.

Several other references have also indicated that the vast majority of root-zone establishment is concentrated in the top layer of rooting media (US EPA, 1993; Reed et al., 1995). For example, Parr (1990) surveyed 12 HSSF TWs in the U.K. and found that 70–100% of root growth occurred in the top 20 cm of the media. It is believed that this vertical gradient in plant establishment has a detrimental effect on the hydraulic efficiency of HSSF TWs. Numerous researchers have performed internal hydrodynamic visualizations using water tracers and report similar observations of preferential flow along the bottom of the bed (Spangler et al., 1976; Gersberg et al., 1984; Bowmer, 1987; Fisher, 1990; Waters et al., 1993; Breen and Chick, 1995; García et al., 2003; Knowles et al., 2010). It has been postulated that the higher hydraulic resistance in the vegetated upper gravel layer causes flow to short-circuit along the less resistive bottom layer.

Some authors have found that water retention in the litter layer and contribution of plant evapotranspiration counteracts any discernible loss of hydraulic retention time due to plant-mediated clogging. For example, when studying planted and unplanted HSSF TW mesocosms in Canada, Chazarenc et al. (2007) found that planted systems had consistently longer HRTs than unplanted systems. One possible benefit of plants to SSF TW hydrology is mitigating the surface flow associated with surface clogging. This occurs due to the wind-induced sway of emergent shoots and stems that disturb the surface sludge layer and create macro-pores through to the subsurface. It is perceived that this mechanism is particularly beneficial to the operation of VF TWs (Brix, 1994a; Molle et al., 2006).

#### 2.4. Chemical effects

Chemical treatment processes, such as adsorption and precipitation, can also contribute to clogging. Physico-chemical adsorption is associated with the removal of metals, petroleum, and synthetic hydrocarbons, ammoniacal-nitrogen, and phosphorous (Wallace and Knight, 2006). For instance, it is common practice for HSSF TWs in Norway to incorporate Lightweight Expanded Clay Aggregates with high sorption capacity, for the purposes of phosphorous removal (Zhu et al., 1997). In general, adsorbate films do not become thick enough to create clogging problems as the adsorption is limited by the sorption capacity of the media.

However, chemical precipitation of metal hydroxides and sulphides (Sheoran and Sheoran, 2006), calcium carbonate (Fleming et al., 1999) and elemental sulphur (Kadlec and Wallace, 2009) may form film-like coatings on media surfaces. Chemical transformations within a wetland environment can be encouraged through biological interactions (Dotro et al., 2010), and systems that treat industrial wastewaters may clog in this manner if both the rate of biological activity and concentration of these compounds is high (Kadlec and Wallace, 2009). Such a phenomenon was described by Nivala et al. (2007), who observed severe iron fouling of the media in an aerated HSSF TW for the treatment of landfill leachate. Nivala et al. (2007) attributed this to ferric hydroxide precipitation under the highly oxidizing conditions, which then sorbed to media surfaces and associated with interstitial biological matter such that porosity was heavily reduced and media became welded together.

#### 2.5. Clog matter composition

The biodegradability of accumulated clog matter dictates its impact on clogging. Initially it was assumed that organic matter would decompose sufficiently such that only inorganic solids would contribute to SSF TW clogging. Based on an influent inorganic solids loading of  $8 \text{ kg/m}^3$  over 18 months, US EPA (1993) predicted that only 1% of interstitial volume would be lost from a HSSF TW. Similar calculations based on the assumption of clogging solely by inorganic matter were responsible for early overestimations regarding system longevity (Conley et al., 1991; Bavor and Schulz, 1993; Blazejewski and Murat-Blazejewska, 1997). However, the literature confirms that clogging can be caused by accumulation of both organic and inorganic matter.

Regarding predominantly organic clog matter: Tanner et al. (1998) state that 80% of accumulations in a dairy wastewater HSSF TW were volatile although subsequent investigation found that 63–96% of these organic matter fractions were relatively refractory (Nguyen, 2000, 2001). Accumulations were predominantly composed of humic, humin and fulvic acids, derived from lignocellulosic and humic compounds in the dairy wastewater and plant detritus. Nguyen (2000) discusses that variable decomposition rates would lead to a mixture of macro-morphological plant

residues and morphologically unstructured humic compounds with a range of molecular weights and packing densities. Furthermore, humic compounds are highly colloidal and amorphous with great hydrophilic potential and physical binding properties (Christensen et al., 1998; Nguyen, 2000). Indeed, the accumulation of clog matter in this site was approximately 80% water by volume (Tanner and Sukias, 1995). These properties give humic organic matter great potential for pore blockage.

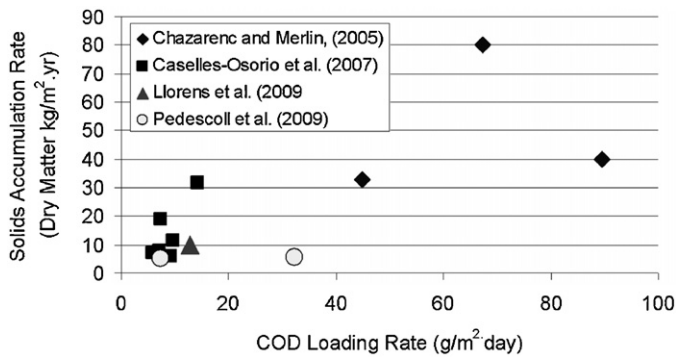
Regarding predominantly inorganic clog matter accumulations: Platzer and Mauch (1997) reported that less than 3% of the deposits in a soil-based VF TW in Merzdorf, Germany were organic. Kadlec and Watson (1993) described that over 80% of the clog matter accumulated in a HSSF TW for lagoon effluent polishing was inorganic in composition, explaining that biological matter was being satisfactorily degraded. Similarly, Caselles-Osorio et al. (2007), Llorens et al. (2009) and Pedescoll et al. (2009) also reported clog matter accumulations with inorganic content greater than 75% in six secondary treatment HSSF TWs in Catalonia, Spain. From the information presented by Llorens et al. (2009) and Kadlec and Watson (1993) the porosity loss attributable to accumulation of inorganic solids in each of these studies is very small, and in agreement with earlier calculations mentioned above (1–3% pore occlusion based on an inorganic solids density of  $2650 \text{ kg/m}^3$ ). However, complexation with the small quantities of biological material present can form low-density gelatinous sludge with very high water retention capacity (Kadlec and Watson, 1993; Tanner et al., 1998; IWA, 2000). The composite clog matter in the studies of Llorens et al. (2009) and Kadlec and Watson (1993) had resulting densities between 33 and  $371 \text{ kg/m}^3$  which increased effective pore occlusion to between 27% and 77%. In the case of Kadlec and Watson (1993) the authors postulate that this occurred due to silica algae combining with calcium and aluminium at low redox values. In the case of Llorens et al. (2009), the sludge density in the inlet region was on average four times lower than that at the outlet region ( $60 \text{ kg/m}^3$  versus  $240 \text{ kg/m}^3$ ) but was doubly effective at reducing porosity (66% occlusion versus 28%). This corresponded with lower values of hydraulic conductivity at the front end of the bed (20 m/d versus 45 m/d) (Pedescoll et al., 2009).

### 3. Design and operational influences

The distribution of clogging through the system and the speed with which it develops depends on numerous design and operational factors, such as the wastewater loading rate and pollutant characteristics, the physical properties of the porous media, inlet arrangement and system dimensions, and whether the system is operated with periods of resting. This section elaborates on the influence of these parameters.

#### 3.1. Wastewater characteristics

Knowledge of wastewater characteristics, such as flow rates, solids content and pollutant characteristics, are vital for understanding clogging processes in SSF TWs. In both HSSF TWs and VF TWs, hydraulic overloading may lead to periods of overland flow. This increases the likelihood that surface layer accumulation rates will exceed mineralization rates and intervention may be required to prevent complete hydraulic failure (Platzer and Mauch, 1997; Langergraber et al., 2003; Zhao et al., 2004). For example, Molle et al. (2006) established that French VF treatment wetlands will work reliably as long as they are not consistently hydraulically overloaded. Systems that had been consistently overloaded were found to have poorly mineralized surface sludge deposits with infiltration rates as low as 2.5 m/d, thus providing a bottleneck to flow.



**Fig. 1.** Correlation between areal COD loading rate and solids accumulation rate for three VF TWs in France (Chazarenc and Merlin, 2005) and six HSSF TWs in Catalonia (Caselles-Osorio et al., 2007; Llorens et al., 2009; Pedescoll et al., 2009).

Regarding pollutant characteristics; both the physical form and biodegradability of wastewater constituents will affect clogging. Low TSS does not necessarily preclude clogging, as demonstrated by Caselles-Osorio and García (2006) who performed side-by-side tests on experimental HSSF TWs system fed with different forms of organic matter (glucose versus starch). Their results suggest that the system fed with dissolved glucose had lower inlet hydraulic conductivities than a system fed with particulate starch, leading them to conclude that glucose stimulated greater biofilm clogging due to its readily biodegradable nature.

In general, most authors report a positive correlation between system clogging and both TSS and COD loading rates (Tanner and Sukias, 1995). Winter and Goetz (2003) found loading rates of TSS and COD correlated positively with severity of clogging in a survey of 21 VF TWs in Germany. Clogging problems were not apparent in systems that received less than 20 g/m² d COD and 5 g/m² d TSS.

Building on the work of Caselles-Osorio et al. (2007), Fig. 1 compares the COD loading rate versus solids accumulation rates for six HSSF TWs in Catalonia, Spain (Caselles-Osorio et al., 2007; Llorens et al., 2009; Pedescoll et al., 2009), and three VF TWs in France (Chazarenc and Merlin, 2005). Over two years, the secondary HSSF TW systems in Catalonia received average influent COD loading rates of 12 g/m² d with a resulting average solids accumulation rate of 11 kg/m² yr. Accordingly, the primary VF TW of Chazarenc and Merlin (2005) received a larger COD loading rate (67 g/m² d on average) corresponding to a proportionally greater solids accumulation rate (51 kg/m² yr on average). Despite this the French systems purportedly operated without clogging whereas evidence of surface ponding existed in several of the Catalanian HSSF TWs.

### 3.2. Intermittent operation

Intermittent operation involves feeding wastewater in a manner such that subsurface aeration will occur, which restores aerobic conditions and accelerates clog matter mineralization. This principle is fundamental to the operation of VF TWs. The periodicity of loading to resting will determine the ability of the system to operate without clogging (Cooper, 2005). The required recovery period for a particular cell will depend on climatic conditions. Systems in cold and wet climates will require a longer recovery period than those in hot and arid climates. Resting periods on the order of days to weeks are suggested for VF TWs in Northern Europe (Platzer and Mauch, 1997; Langergraber et al., 2003; Green et al., 2006), with different cells rested in rotation so that plant operation will be uninterrupted.

Variants of VF TWs such as fill-and-drain wetlands (Zoeller and Byers, 1999; Lahav et al., 2001), tidal flow wetlands (Zhao et al., 2004; Austin, 2005) and reciprocating wetlands (Behrends et al., 2001) incorporate operating strategies that promote increased subsurface oxygen availability, such as water level fluctuation, intermittent dosing, and alternate operation. Additionally, these systems operate without the typical application of wastewater over the surface of the bed, which in principle reduces the tendency of the system to clog on the surface. Zhao et al. (2004) showed that clogging of a tidal flow wetland could be mitigated by arranging gravel fractions with increasing diameter from top to bottom, so that accumulated solids were better distributed throughout the reactor depth. However, studies indicate that these systems may still clog in the upper surface despite their operation mode (Cooper and Cooper, 2005; Sun et al., 2007).

Studies also suggest that intermittent operation may be beneficial for reversing clogging in HSSF TWs (Nguyen, 2000; Corzo et al., 2008). Batchelor and Loots (1997) rested a field-scale HSSF TW to allow for surface sludge layer mineralization, and successfully restored infiltration to the subsurface.

### 3.3. Media characteristics

The hydraulic conductivity of a porous media is very sensitive to media size, as emphasized by the squared relationships in the Kozeny-Carmen equation for flow through porous media (Kadlec and Knight, 1996). Despite this, smaller media, such as soil and sands, were originally used (Brix and Schierup, 1989; Coombes, 1990; Haberl and Perfler, 1990; Netter and Bischofsberger, 1990) as they were believed to offer superior treatment performance to larger media. Indeed, the smaller the particle size, the higher specific surface area available for biofilm establishment, and surface chemistry; and the greater the likelihood of suspended solids interception due to narrower pore diameters. However, this makes fine media prone to rapid clogging by pore occlusion from filtration and the bridging of surface accumulations (Blazejewski and Murat-Blazejewska, 1997; Platzer and Mauch, 1997; Langergraber et al., 2003; Wallace and Knight, 2006).

Subsequent work by Griffin et al. (2008) on the performance of 25 HSSF TWs in the U.K. showed that larger media, with median diameters between 6 and 11 mm, did not impair the ability of the system to achieve treatment requirements. All media sizes achieved effluent BOD levels below 5 mg/L, although according to the Kozeny-Carmen equation, the 11 mm substrate would have a clean hydraulic conductivity over three times greater than the 6 mm substrate. Resultantly, the desire to maximize asset longevity has led to a gradual increase in the media sizes employed in HSSF TWs and towards the implementation of coarser media, such as gravels (Cooper et al., 1996; US EPA, 2000; Iwema et al., 2005; ÖNORM, 2005; García and Corzo, 2008) as reflected in Table 3 which summarizes media specifications that have been published over the years (Wallace and Knight, 2006; Kadlec and Wallace, 2009).

Particle size distribution and particle shape also influence media hydraulic conductivity. Adaptations of the Ergun equation are available to describe the influence of grain size distribution on bulk media hydraulic conductivity (Kadlec and Knight, 1996). Media that is relatively non-spherical or angular will exacerbate clogging as it reduces porosity and increases the specific surface area available for biofilm growth (Kadlec and Knight, 1996; Hyánková et al., 2006). The significant benefit to asset lifetime offered by controlling media specification was emphasized by Griffin et al. (2008), who announced that Severn Trent Water (a U.K. water utility) HSSF TWs built after 2008 would only incorporate washed round gravels with a narrow distribution of particle sizes between of 10–12 mm.

**Table 3**

Examples of media size distributions recommended by various national and international design guidelines and publications. Adapted from Wallace and Knight (2006) and Kadlec and Wallace (2009).

| Country                         | Size distribution                                                                | Source                                                                                                           |
|---------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Austria                         | 0–4 mm (gray water)<br>1–4 mm (tertiary treatment)<br>4–8 mm (primary treatment) | ÖNORM (2005)                                                                                                     |
| Czech Republic                  | $d < 20$ mm                                                                      | Vymazal (1996)                                                                                                   |
| Germany                         | 0.2–1.0 mm (sand)                                                                | GFA (1998)                                                                                                       |
| United Kingdom                  | 10–12 mm                                                                         | Griffin et al. (2008)                                                                                            |
| United States                   | 1–8 mm<br>3–6 mm<br>12–25 mm<br>2–28 mm<br>20–30 mm<br>$d > 4$ mm                | US EPA (1988)<br>TVA (1993)<br>US EPA (1993)<br>Reed et al. (1995)<br>US EPA (2000)<br>Wallace and Knight (2006) |
| European Design Guidelines      | 3–6 mm<br>6–12 mm                                                                | EC/EWPCA (1990)                                                                                                  |
| International Water Association | 8–16 mm                                                                          | IWA (2000)                                                                                                       |

### 3.4. Upstream treatment processes

Accidental spillover of solids and sludge from upstream treatment processes can create shock loads of solids to downstream SSF TWs (Cooper et al., 2005; Caselles-Osorio et al., 2007). Good design and maintenance of upstream processes can mitigate the likelihood of this occurrence; however, it is recommended that effluent filters or clarifiers are used between the upstream process and the SSF TW to minimize the impact of solids carry-over (Tchobanoglous, 2003).

Alternative pretreatment technologies are under investigation for use in conjunction with SSF TWs, with the aim of minimizing solids loading and thus, maximizing system lifetime. Caselles-Osorio and García (2007) estimate that when compared against primary-settled effluent, physicochemical pre-treatment could extend the life of a HSSF TW by approximately 10 years. They note, however, that the cost and O&M requirements of coagulant, energy usage and sludge handling responsibilities will prevent this option from being suitable in every situation. Other possible pretreatments include Upflow Anaerobic Sludge blanket (UASB) reactors (Green et al., 2006; Barros et al., 2008; Dornelas et al., 2008), Hydrolytic Upflow Sludge Bed reactors (HUSB) (Álvarez et al., 2008; Corzo et al., 2008) and floating treatment wetlands (Griffin et al., 2008); all of which can produce a primary effluent relatively low in COD and TSS in comparison to conventional settling. However, some of the aforementioned technologies, such as UASB, are best suited to warm climates.

### 3.5. System aspect ratio and inlet distribution

As previously explained, the factors responsible for clogging lead to preferential clog matter accumulation in the vicinity of the inlet, where the wastewater is most concentrated. The extent of preferential clogging can be mitigated by uniformly loading the wastewater over the greatest area possible, and is hence controlled by the system dimensions and the influent distribution arrangement. In VF TWs this is achieved by designing for uniform areal distribution. In HSSF TWs, early designs had aspect ratios with greater lengths than widths, which helped to achieve good width distribution but concentrated the cross-sectional loading of wastewater (Watson et al., 1990). Resulting experience of extensive clogging and overland flow in the inlet region of these systems prompted the Danish to modify the aspect ratio, so that the

width exceeded the length (Brix and Schierup, 1989). This design reportedly suffered less overland flow issues and hence became recommended by most HSSF TW design guidelines.

Regarding influent distributor design, the operation of typical VF TWs necessitates distributors that are located on or near the surface of the bed so that vertical percolation through the media will occur. Uniform areal distribution is approached using a network of perforated pipes that cover the majority of the surface, or by using concrete splash pads (GFA, 1998; Brix and Arias, 2005; Molle et al., 2005; ÖNORM, 2005).

In HSSF TWs, whether clogging initially develops on the surface or within the subsurface corresponds to whether a surface or subsurface-based influent distributor is used. Experience in the U.K. has shown that poorly designed, incorrectly installed or ill-maintained inlet distributors will encourage uneven distribution of the flow (Rousseau et al., 2005; Griffin et al., 2008) and hence uneven clogging of the bed media (Knowles et al., 2010). Early inlet distributor designs in U.K. HSSF TWs exacerbated this situation, as they generally comprised discrete point sources, distributed at multiple points along the width of the bed, such that flow was not loaded uniformly (Murphy and Cooper, 2010). Cooper et al. (2008) found inlet distribution problems that required intervention at 34 from a survey of 255 HSSF TWs in the U.K. In the U.K., the design of surface based distributors for HSSF TWs has evolved to try and achieve better width distribution, for instance by using troughs that span the width of the bed (Murphy and Cooper, 2010). Troughs are still susceptible to solids accumulation but are preferred by operators because of their ease of maintenance (Griffin et al., 2008).

Some early U.S. HSSF TWs also utilized surface manifolds with adjustable multiple outlet ports (Steiner and Freeman, 1989; Watson and Hobson, 1989). However, the design of these systems eventually shifted towards the use of subsurface manifolds (TVA, 1993; US EPA, 1993). Initial subsurface distributors for HSSF TWs comprised a pipe distributor, located a few centimeters below the gravel surface. Designs of this type are still prevalent in mainland Europe (Vymazal et al., 1998; Vymazal and Kröpfelová, 2008), but they have been superseded in the U.S. by subsurface infiltration chambers perpendicular to the flow direction (Campbell and Ogden, 1999; Wallace and Knight, 2006), due to the opinion that simple pipe distributors resulted in clogging in the vicinity of the pipe. The use of subsurface infiltration chambers is intended to maximize the cross-sectional area perpendicular to the direction of flow over which the solids are loaded and reduce preferential accumulation of solids. To further minimize cross-sectional organic loadings, some U.S. authors have utilized extended influent distributors that load wastewater along the width of the bed, but also include distribution manifolds that protrude longitudinally into the cell (Muñoz et al., 2006; Wallace and Knight, 2006).

Another method for achieving uniform width distribution has been to incorporate an open trench at the inlet that precedes subsurface flow. This also serves as a solids retention basin, thus reducing accumulation of clog matter within the gravel (US EPA, 1993; King et al., 1997; Murphy and Cooper, 2010).

## 4. The influence of clogging on SSF treatment wetland hydraulics

Problems associated with clogging are usually hydraulic malfunctions, such as undesirable ponding of wastewater on the surface of the system. However, the motivation to intervene may arise for different reasons. For example, VF TWs typically operate with impulse dosing of wastewater onto the surface. As clogging develops, the infiltration time for each does will increase such that wastewater begins to pond on the surface, until the ponded water

**Table 4**

Reported values for media hydraulic conductivities in the inlet and outlet regions of 21 field-scale HSSF TWs. Information is also included regarding the design and operation of these systems. Adapted from Knowles et al. (2010).

| Reference                     | System name     | Treatment type    | Hydraulic loading rate (mm/d) | Dimensions L:W:D (m) | Age at time of study (months) | Substrate size range (mm) | Inlet conductivity (m/d) <sup>a</sup> | Outlet conductivity (m/d) <sup>a</sup> | Method of measurement |
|-------------------------------|-----------------|-------------------|-------------------------------|----------------------|-------------------------------|---------------------------|---------------------------------------|----------------------------------------|-----------------------|
| Caselles-Osorio et al. (2007) | Verdú 1         | Secondary         | 190                           | 30:31:0.5            | 48                            | 6–12                      | 2                                     | 12                                     | Falling head          |
|                               | Verdú 2         | Tertiary          | 409                           | 27:16:0.4            | 48                            | 6–12                      | 25                                    | 61                                     | Falling head          |
|                               | Alfés           | Secondary         | 49                            | 32:38:0.5            | 48                            | 6–12                      | 7                                     | 2                                      | Falling head          |
|                               | Corbins         | Primary           | 178                           | 35:35:0.5            | 48                            | 6–12                      | 2                                     | 200                                    | Falling head          |
|                               | Almatret N      | Secondary         | 54                            | 23:20:0.5            | 36                            | 6–12                      | 1                                     | 87                                     | Falling head          |
|                               | Almatret S      | Secondary         | 54                            | 28:18:0.5            | 36                            | 6–12                      | 1                                     | 82                                     | Falling head          |
| Kadlec and Watson (1993)      | Benton Cell 3   | Secondary         | 17                            | 333:44:0.8           | 41                            | 14–23                     | 2500                                  | 27,500                                 | Survey                |
| Watson and Choate (2001)      | Jones           | Secondary         | 10                            | 13:3.1:0.5           | 72                            | 3–6                       | 1000                                  | 5400                                   | Survey                |
|                               | Gray            | Secondary         | 19                            | 12:3:0.5             | 72                            | 3–6                       | 10,200                                | 8100                                   | Survey                |
|                               | Terrell         | Secondary         | 21                            | 2.7:1.8:0.3          | 72                            | 3–6                       | 4,900                                 | 4700                                   | Survey                |
|                               | Snelling        | Secondary         | 58                            | 4.3:2.4:0.3          | 72                            | 3–6                       | 85                                    | 325                                    | Survey                |
| Fisher (1990)                 | <i>Scirpus</i>  | Secondary         | 50                            | 100:4:0.5            | 33                            | 3–10                      | 1800                                  | 25,000                                 | Survey                |
|                               | <i>Typha</i>    | Secondary         | 48                            | 100:4:0.5            | 33                            | 3–10                      | 2500                                  | 25,000                                 | Survey                |
|                               | Control         | Secondary         | 38                            | 100:4:0.5            | 33                            | 3–10                      | 2500                                  | 25,000                                 | Survey                |
| Sanford et al. (1995)         | Bed 4           | Landfill leachate | 14                            | 33:3:0.6             | 26                            | 5                         | 4150                                  | 3370                                   | Survey                |
| Knowles et al. (2010)         | Moreton Morrell | Tertiary          | 467                           | 15:15:0.6            | 177                           | 3–9                       | 2                                     | 26,000                                 | Constant head         |
| Knowles and Davies (2009)     | Fenny Compton   | Tertiary          | 171                           | 12:40:0.6            | 12                            | 6–12                      | 1065                                  | 84,000                                 | Constant head         |
| Drury and Mainzhausen (2000)  | Cell 1          | Acid drainage     | 41                            | 33:33:0.7            | 30                            | 6–20                      | 6                                     | 3500                                   | Survey                |
|                               | Cell 2          | Acid drainage     | 100                           | 16:28:1.2            | 30                            | 6–20                      | 6                                     | 3500                                   | Survey                |
| Pedescoll et al. (2009)       | Verdú 1         | Secondary         | 77                            | 30:31:0.6            | 177                           | 6–12                      | 20                                    | 45                                     | Falling head          |
|                               | Corbins         | Secondary         | 59                            | 35:35:0.5            | 218                           | 6–12                      | 3                                     | 55                                     | Falling head          |

<sup>a</sup> denotes average value across the width of the bed.

no longer completely infiltrates between doses (Molle et al., 2006; Sun et al., 2007). However, it is the associated treatment problems that this causes, rather than the hydraulic issues themselves, which necessitate intervention.

Contrastingly, intervention into clogged HSSF TWs is usually prompted by the need to correct the hydraulic problem. The extent of ponding that qualifies as “undesirable” varies depending on operator expectations and regulatory requirements. For example, U.S. regulatory agencies prohibit the surface exposure of wastewater to protect public health (US EPA, 2002) and therefore HSSF TWs exhibiting even a small degree of surface flow will necessitate restorative action (Kadlec and Wallace, 2009). A much greater degree of ponding is often tolerated in U.K. HSSF TWs before the problem is addressed (Cooper et al., 2005), as these systems are typically situated within enclosed wastewater treatment plants. Declining treatment performance has rarely been cited as motivation for restoration in clogged HSSF TWs, as supported by Rousseau et al. (2005) who found that twelve treatment HSSF wetlands in the U.K. all managed to consistently meet discharge standards despite exhibiting symptoms of heavy clogging.

Surface flow most often occurs in the inlet region of SSF TWs, where clogging is greatest due to the previously discussed factors. In HSSF TWs, a corresponding longitudinal gradient in hydraulic conductivity has been measured by numerous authors, with values at the inlet at least an order of magnitude lower than values at the outlet (Table 4). For example, for the 21 systems listed in Table 4 which all have varying design and operational parameters, the measured inlet hydraulic conductivity was on average 60% lower than the outlet hydraulic conductivity.

Numerous guidelines have been proposed to try and help designers account for clogging in system sizing, and to help operators maximize longevity. These often take the form of areal loading

rates for total solids, biochemical and chemical oxygen demand (Table 5), although Wallace and Knight (2006) use the cross-sectional area in the direction of flow to recommend a maximum loading rate of 250 g/m<sup>2</sup> d BOD for HSSF TWs. Table 5 also lists early guidelines for equilibrium hydraulic conductivity, which represents the equivalent bulk reactor hydraulic conductivities that best describe the total head loss observed across field scale HSSF TWs. These guidelines were prescribed on the assumption that clogged hydraulic conductivity would eventually reach an equilibrium determined by the balance between plant growth and clog matter accumulation (Cooper et al., 1996). However, these guidelines do not consider spatial variations in hydraulic conductivity and are generally unable to describe the low inlet hydraulic conductivities reported in Table 4. Resultantly, they have been unsuccessful at producing HSSF TW designs that operate without clogging. Further, Knowles et al. (2010) highlighted the specificity of these guidelines by comparing the suggested equilibrium hydraulic conductivity guidelines for 5 mm gravels, finding values that varied between 86 and 2600 m/d.

## 5. Clogging in common variants of SSF treatment wetlands

The consolidated literature confirms that the development of clogging is highly specific to system design and operation (e.g., horizontal or vertical flow; surface or subsurface inlet loading). Considering the material presented so far it is possible to differentiate how clogging develops in common SSF TW design variants, based on the relationship between flow configuration, design features, and operational parameters. Here we make a distinction between HSSF TWs with surface or subsurface influent distribution, and separate discussion is given to VF TWs with sand or gravel media.

**Table 5**  
Published guidelines that have been suggested for the design consideration and prevention of clogging in various SSF treatment wetland designs. These include equilibrium hydraulic conductivity of media in SST treatment wetlands that will result from clogging, and loading limits for hydraulic loading rate (HLR), total suspended solids (TSS) and biological oxygen demand (BOD).

| Type | Treatment | Location       | Media type and size         | Equilibrium hydraulic conductivity (m/d) | BOD loading limit (g/m <sup>2</sup> d) | COD loading limit (g/m <sup>2</sup> d) | TSS loading limit (g/m <sup>2</sup> d) | Reference                       |
|------|-----------|----------------|-----------------------------|------------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|---------------------------------|
| HSSF | All       | United Kingdom | Gravel (3–6 mm)             | 86                                       | –                                      | –                                      | –                                      | Cooper et al. (1996)            |
| HSSF | Secondary | United States  | Gravel (3–6 mm)             | 2600                                     | –                                      | –                                      | –                                      | TVA (1993)                      |
| HSSF | Secondary | Germany        | Soil/sand (0.2–1.0 mm)      | 78                                       | –                                      | –                                      | –                                      | GFA (1998)                      |
| HSSF | All       | Europe         | Gravel (8–16 mm)            | 1782                                     | –                                      | –                                      | –                                      | IWA (2000)                      |
| HSSF | Primary   | Austria        | Gravel (4–8 mm)             | 9–173                                    | –                                      | –                                      | –                                      | ÖNORM (2005)                    |
| HSSF | Secondary | Australia      | Gravel (5–10 mm)            | –                                        | –                                      | –                                      | 40                                     | Bavor and Schulz (1993)         |
| HSSF | Secondary | South Africa   | Gravel (35 mm)              | –                                        | –                                      | –                                      | 42                                     | Batchelor and Loots (1997)      |
| HSSF | Secondary | United States  | Gravel ( $d_{10} > 4$ mm)   | 260                                      | 250 <sup>a</sup>                       | –                                      | –                                      | Wallace and Knight (2006)       |
| HSSF | Secondary | United States  | Gravel (20–30 mm)           | 1000                                     | 6                                      | –                                      | –                                      | US EPA (2000)                   |
| VF   | Secondary | Austria        | Sand (0.06–4 mm)            | –                                        | –                                      | –                                      | 20                                     | Langergraber et al. (2003)      |
| VF   | Primary   | France         | Graded gravels (0–70 mm)    | –                                        | 30–35                                  | –                                      | –                                      | Liénard et al. (1990)           |
| VF   | Secondary | Germany        | Sand ( $d_{10}$ 0.1–0.3 mm) | –                                        | –                                      | 20                                     | 5                                      | Winter and Goetz (2003)         |
| VF   | Secondary | Germany        | Sand ( $d_{10}$ 0.1–0.2 mm) | –                                        | –                                      | 25                                     | –                                      | Platzer and Mauch (1997)        |
| VF   | Secondary | United States  | Gravel ( $d > 4$ mm)        | –                                        | 10–40                                  | –                                      | –                                      | Crites and Tchobanoglous (1998) |

<sup>a</sup> Refers to cross-sectional loading (perpendicular to direction of flow). All other loadings are areal-based.

### 5.1. Horizontal subsurface flow treatment wetlands with surface loading

Systems of this nature are popular in the U.K. where they are widely used for the tertiary treatment of municipal wastewater (Cooper, 2007); however, reports of clogging in similar systems have also emanated from France (Liénard et al., 1990), Spain (Caselles-Osorio et al., 2007; Llorens et al., 2009; Pedescoll et al., 2009) and South Africa (Batchelor and Loots, 1997). In U.K. systems, the upstream processes are typically rotating biological contactors or trickling filters (Griffin and Pamplin, 1998), which are prone to sloughing solids to the downstream TW. The wastewater is continuously fed through surface-based inlet distributors that span the width of the bed (Fig. 2). Distributor designs vary and include pipes with multiple risers and troughs with numerous distribution weirs. Historically, the beds have employed gravels on the order of 6–12 mm. Other details regarding the design of this system are well described in Cooper et al. (1996).

The nature of wastewater loading onto these systems results in preferential clogging at the inlet, within the upper layers of gravel and on the surface. Clog matter formations are usually a composite of wastewater solids and patchy masses of leaf litter in various stages of decomposition (Tanner et al., 1998), which are prevented from adequately mineralizing by overland flow of wastewater. For example, studies of the dairy wastewater systems of Tanner et al. (1998) identified that the accumulated clog matter manifested as a 50 mm-deep sludge layer and penetrated into the top 100 mm of media (Nguyen, 2001). In their survey of 255 HSSF TWs in the U.K., Cooper et al. (2008) frequently encountered systems with surface sludge accumulations in excess of 150 mm at the inlet and 40 mm at the outlet. Rousseau et al. (2005) made similar observations in their survey of 12 HSSF stormwater treatment wetlands in the U.K., reporting that the vast majority of them had experienced sludge build-up over the entire surface of the bed. These authors also speculate that symptoms such as poor reed growth and weed infestation may be connected to clogging (Cooper et al., 2005, 2008; Rousseau et al., 2005).

Overland flow results across the surface layer unto the point where cumulative infiltration sufficiently meets the HLR. In 30% of the 255 systems surveyed by Cooper et al. (2008), overland flow was present over the majority of the HSSF TW surfaces. Resultantly, these systems may begin operating with vertical flow in addition to the intended horizontal flow regime (Knowles et al., 2010). The

pervasiveness of these problems in U.K. systems is highlighted by frequent incorporation of an overflow pipe during construction. This prevents overspill incidents in extensively clogged systems where the hydraulic loading rate would exceed the diminished conveyance capacity of the subsurface (Fig. 2).

### 5.2. Horizontal subsurface flow with subsurface loading

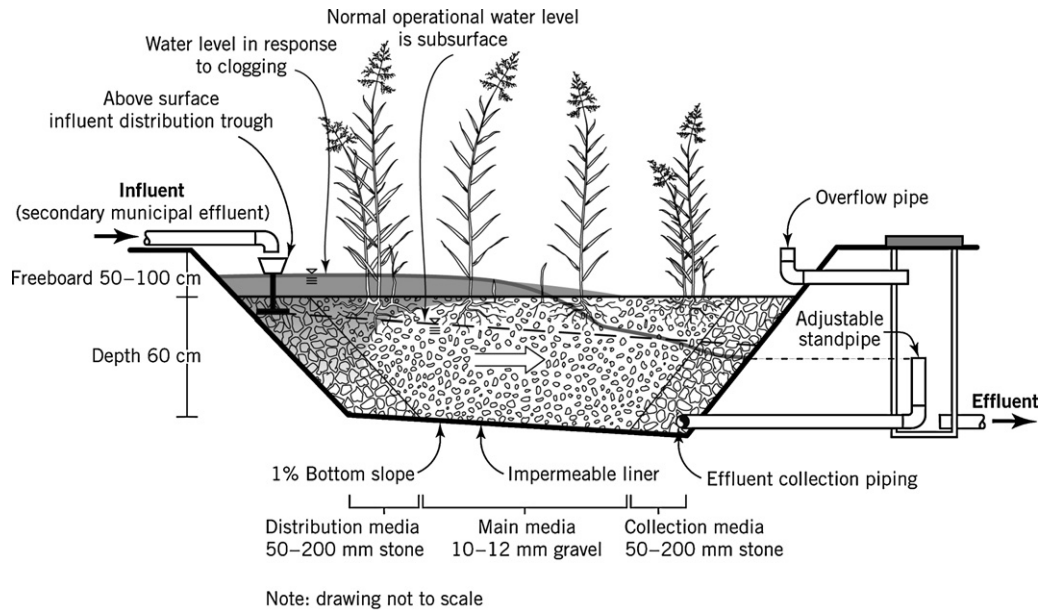
Systems of this nature are popular in the U.S., Australia and parts of continental Europe where they are predominantly used for the secondary treatment of domestic wastewater. The wastewater is continuously fed directly into the subsurface through slotted pipes that are located a few centimeters below the gravel surface (Fig. 3). Design guidelines in Europe recommend gravel sizes of 3–16 mm whilst those in the U.S. have suggested inclusion of media as large as 30 mm. Design guidelines relevant to these systems can be found in IWA (2000), US EPA (2000), and Wallace and Knight (2006).

The subsurface loading results in a preferential accumulation of solids in the proximity of the influent distributor. The primary-treated wastewaters received at the inlet often result in anaerobic conditions, as typified by the development of black coatings on the gravel surfaces (Bowmer, 1987; Kadlec and Watson, 1993; Suliman et al., 2006; Wallace and Knight, 2006; Vymazal and Kröpfelová, 2008).

In advanced stages of clogging, the loss of conductivity at the inlet forces the flow to surface and take an overland flow path until it is able to sufficiently infiltrate through the surface layer and back into the subsurface. Reports describing clogging of this nature have come from the U.S. (Watson et al., 1990; Kadlec and Watson, 1993; US EPA, 1993; Zachritz and Fuller, 1993), Australia (Fisher, 1990), Canada (Chazarenc et al., 2007), Czech Republic (Vymazal, 1996) and Poland (Maloszewski et al., 2006). Once flow has surfaced, clogging dynamics similar to those described for surface loaded HSSF TWs may develop.

### 5.3. Vertical flow with sand matrix

The original vertical flow systems of Seidel (1973) incorporated sand or soil substrates, after which these systems became popular in countries such as Germany, Austria, and Denmark where they are commonly used for the secondary treatment of domestic wastewater. A schematic of a typical VF TW with a sand matrix is shown in Fig. 4. The most common influent distribution sys-



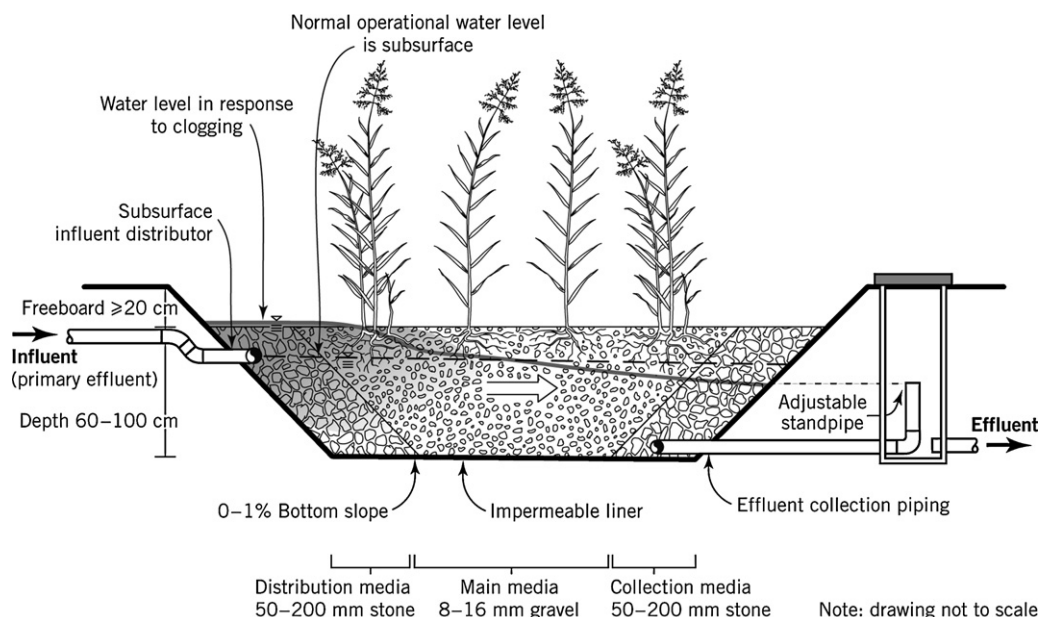
**Fig. 2.** Clogging profile for a typical HSSF wetland with above-surface influent distribution. Design details are adapted from Griffin et al. (2008); clogging profile is based on information from Knowles et al. (2010).

tem consists of perforated piping, equally distributed to cover the surface area of the bed. The first layer of media is generally consists of 8–16 mm gravel to enhance flow percolation throughout the treatment bed. The main treatment media is a mixture of sand and fine gravel with an average grain size smaller than 4 mm. Below the main treatment layer, a transition layer and a drainage layer of increasingly coarser gravels is used to facilitate final effluent collection. Wastewater dosing is typically intermittent as this allows greater oxygen transfer within the wetland bed and promotes aerobic degradation processes. Further details regarding the specification of these systems are given in ATV-A 262 (GFA, 1998), Brix and Johansen (2004) and ÖNORM (2005).

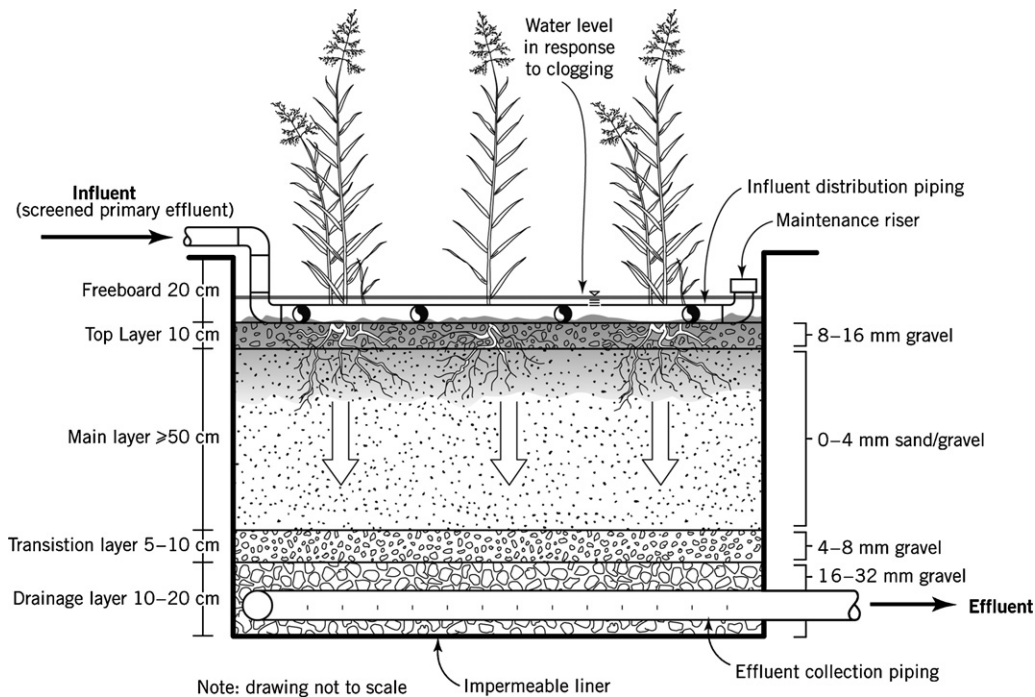
The use of sand as opposed to gravel means that these systems are highly prone to clogging if they are operated incorrectly (Cooper

and Green, 1995). For example, Winter and Goetz (2003) reported that six from a survey of 21 VF filters in Germany had clogged and found that this was mainly linked to overloading of wastewater. Clogging preferentially develops in the top part of the main layer, where the greatest biological growth occurs due to contact between the fine media, nutrients and organic matter in the influent wastewater. Indeed, Tietz et al. (2007) found through the use of four different methods that microbial biomass was most abundant in the top 10% of a VF TW filter.

Clogging can be extremely rapid if the period between doses does not allow sufficient wastewater percolation and organic matter mineralization (Langergraber et al., 2003), because sealing of the surface by wastewater inhibits further transfer of oxygen for mineralization (Platzer and Mauch, 1997). Kayser



**Fig. 3.** Clogging profile for a typical HSSF wetland with subsurface influent distribution. Design details are adapted from Vymazal et al. (1998); clogging profile is adapted from Kadlec and Wallace (2009).



**Fig. 4.** Clogging profile for a typical VF treatment wetland with sand substrate. Design details are adapted from ÖNORM (2005); clogging profile is based on information in Langergraber et al. (2003).

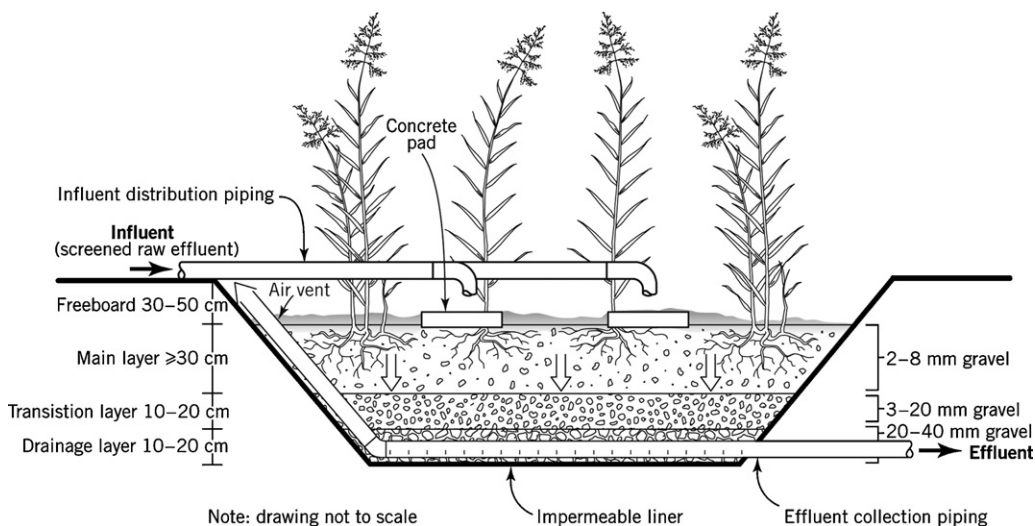
and Kunst (2005) observed that the oxygen content of air in a “healthy” filter consistently returned to the atmospheric level of 21% between doses. However, the oxygen content of air in a clogged filter was below 5% and was relatively unaffected by dosing cycles.

#### 5.4. Vertical flow with gravel matrix

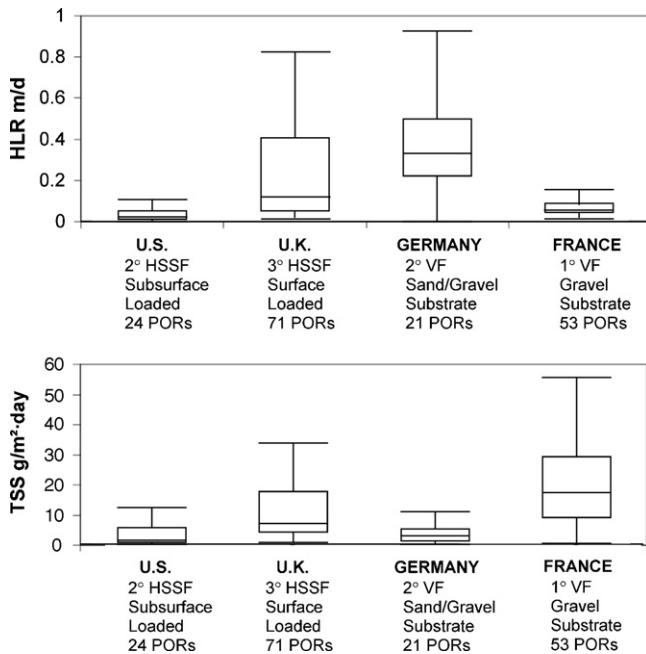
The gravel matrix VF TWs commonly employed in France is used to treat screened municipal wastewaters, and form the first stage of a multiple-cell wetland network. A schematic of the French VF treatment wetland system is shown in Fig. 5. Wastewater is loaded at numerous points over the surface of the bed onto concrete splash

pads that aid areal distribution. The main filter layer is small gravel with a size distribution of 2–8 mm. This layer is followed by a transition layer and a drainage layer with respective gravel sizes of 3–20 and 20–40 mm. Further details regarding the design of these systems can be found in Iwema et al. (2005).

The French type of VF treatment wetlands are designed to accumulate a solids layer on the surface of the bed, through which the applied wastewater must percolate (Iwema et al., 2005; Molle et al., 2005); i.e. they are designed to clog (Kadlec and Wallace, 2009). However, the French wetlands are operated in a manner such that the surface layer is beneficial to treatment performance without becoming detrimental to hydraulic performance (Chazarenc and Merlin, 2005). The wastewater is dosed intermittently, which pro-



**Fig. 5.** Clogging profile for a typical VF (French-type) treatment wetland with gravel substrate. These systems are generally designed with several beds in series; the first bed in the series (shown) is constructed with larger gravel and retains most of the solids. Design details are adapted from Lienard et al. (1998); clogging profile is based on information from Molle et al. (2005).



**Fig. 6.** Box-and-whisker plots showing the relative influent HLR (top) and TSS (bottom) loadings derived from four national treatment wetland databases. POR indicates the entire period of record for one system. Data source: [Constructed Wetland Association Wetland Database \(2006\)](#), [WERF Database \(2006\)](#), data adapted from [Winter and Goetz \(2003\)](#), and data adapted from [Boutin et al. \(1997\)](#).

vides time for the surface layer to dewater and mineralize so that hydraulic conductivity is maintained.

It is believed this mode of operation effectively prevents significant ponding, with [Chazarenc and Merlin \(2005\)](#) citing this as the reason that several gravel-based VF systems in France of ages varying up to eight years showed no signs of clogging. [Molle et al. \(2005\)](#) further support the effectiveness of the French design by reporting that only one out of 71 VF systems reviewed had ever required intervention due to clogging. Generally, mineralization ratios of 60% have prevented this phenomenon from occurring in other systems ([Boutin et al., 1997](#)).

#### 5.5. Why does clogging impact some systems worse than others?

From the literature reviewed thus far it appears that some SSF TW variants are more prone to clogging than others. For example, the primary VF TWs in France appear less prone to problematic clogging than tertiary HSSF TWs in the U.K., although this seems counterintuitive. To elucidate this phenomenon, [Fig. 6](#) compares datasets for hydraulic loading rates (in m/d) and TSS loadings (in g/m² d) to each of the SSF TW variants discussed in the previous section. These datasets are: the [Constructed Wetland Association \(CWA\) Wetland Database \(2006\)](#), from which Periods of Record (POR) were taken for 71 different U.K. HSSF Tertiary TWs with surface influent distribution; the [Water Environment Research Federation \(WERF\) Database \(2006\)](#), from which POR were taken for 24 different U.S. HSSF Secondary TWs with subsurface influent distribution; data adapted from [Winter and Goetz \(2003\)](#), from which POR were taken for 21 different German VF TWs with sand matrix; and data adapted from [Boutin et al. \(1997\)](#), from which POR were taken for 53 different French VF TWs with gravel matrix. Data are compared using a box and whisker chart, where the box represents 50% of each dataset with the internal line at the median value and the whiskers extend to the minimum and maximum value in the dataset, excluding any outliers.

What is evident from [Fig. 6](#) is that U.K. HSSF TWs receive a combination of relatively high HLR and TSS load (median HLR = 0.12 m/d and median TSS load = 7 g/m² d) in comparison to the other datasets, where either one or both measures is relatively lower. The U.S. HSSF TWs receive the lowest solids and hydraulic loading rates (median HLR = 0.02 m/d and median TSS load = 2 g/m² d). The German VF TWs receive a high HLR (median = 0.3 m/d) but a relatively low solids load (median = 3 g/m² d). Statistically, the French VF TWs receive the highest TSS load (17 g/m² d) but a comparatively low HLR (0.06 m/d).

As discussed, problematic clogging is most typically qualified by undesirable surface ponding. However, this does not always correspond to the extent of solids accumulation within the systems. As indicated by [Fig. 1](#), French primary VF TWs accumulate clog matter almost five times faster than secondary HSSF TWs in Spain; however, the low HLR of French systems prevents this from presenting a problem. In contrast, U.K. tertiary systems receive a smaller TSS load than French systems but a much higher HLR, which may explain the propensity of U.K. systems to experience clogging problems. Tertiary U.K. HSSF TWs are sized to be seven times smaller than secondary U.K. HSSF TWs that receive equivalent HLR (0.7 m²/PE versus 5 m²/PE, respectively) ([Green and Upton, 1995](#)). The combination of hydraulic overloading and relatively high solids contributions from upstream processes diminishes the longevity of these systems.

A similar discussion can be used to explain why more reports of clogging problems emanate from Germany than from France; i.e., the German secondary VF TWs operate at lower TSS loads but higher HLR than the French systems. This increases the likelihood that clog matter accumulation in German systems will eventually cause hydraulic problems, whereas in French systems the low HLR means that organic solids are able to mineralize well enough to preclude clogging in most cases. This comparison serves as testament to the claims of clog-free operation in French VF TWs ([Molle et al., 2005](#)). However, French VF TWs generally operate with surface exposure of screened sewage and would not be suitable where local regulations prohibit this.

This comparison has emphasized the importance of considering HLR as well as wastewater pollutant loadings when sizing systems for longevity. However, long-term attempts to circumvent clogging problems will require a better understanding of the biological processes responsible for clog matter reduction, and this cannot be understated.

## 6. Conclusions

Two decades of treatment wetland literature have proven that clogging can limit the asset lifetime of SSF TWs and may threaten the widespread feasibility of the technology. What has become evident in this review, however, is that the development and impact of clogging is specific to system design and operation. By comparing four common variants of subsurface flow treatment wetlands (HSSF TWs with surface or subsurface influent distributor, and VF TWs with sand or gravel matrix), it has been possible to elucidate how design and operation influence clogging. According to the literature, tertiary HSSF TWs in the U.K. are more prone to clogging problems than primary VF TWs in France. This is because French VF TWs are hydraulically operated such that high solids accumulation rates do not cause operational issues, and adequate mineralization of surface clog matter can occur between loadings. Conversely, U.K. systems typically operate with overland flow as a result of high HLR, which prevents clog matter mineralization and causes progressive clogging. Consequently, it appears that both the pollutant mass loading and hydraulic loading rates

are imperative considerations when designing SSF TWs for robust operation.

What is still not known, and is a priority for further research, are the actual mechanisms of clogging that link system design and operational parameters to the characteristics of the accumulated clog matter. Clog matter is a combination of inorganic and organic solids from wastewater treatment processes, and biomass contribution from internal wetland functions; and often has physical properties that are more detrimental to hydraulics than its constituent parts. As a result, the volume or mass of clog matter will not necessarily correlate well with corresponding changes in hydraulic conductivity. This lack of understanding is a barrier to the development of simulation tools for SSF TWs (Langergraber, 2008) and requires new experimental techniques to be developed that can elucidate how the nature of clog matter influences hydraulic conductivity.

General points to consider when designing future SSF TWs to operate with reduced clogging problems include:

1. Use of an influent distributor which uniformly loads the wastewater over the maximum area possible.
2. Intermittent dosing of the influent in surface-loaded systems, which will encourage surface layer mineralization.
3. Reduction of unintentional solids spillover from upstream processes.
4. Incorporation of a pre-treatment stage which negates the impact of upstream process upsets.

It is important that SSF TW design continues to evolve with the intention of reducing clogging. This is especially critical if the current trend of process intensification and footprint reduction is to continue. To aid this, the authors recommend that future design guidelines for SSF TW sizing not be based upon the principles of equilibrium hydraulic conductivity, as these values rarely reflect the hydraulic conductivities typically measured in the field.

## Acknowledgements

Paul Knowles and Jaime Nivala gratefully acknowledge the support and mentoring provided by Scott Wallace over the past 10 years to further understanding of wetland clogging. All authors kindly acknowledge Pascal Molle, Clodagh Murphy, and Scott Wallace for their input. Paul Knowles acknowledges the joint funding kindly provided by Severn Trent Water Plc. (U.K.) and a CASE studentship granted by the EPSRC U.K. (ref. 1121 CASE/CNA/06/28). Gabriela Dotro kindly acknowledges the financial support from a Herff Postdoctoral Fellowship at the University of Memphis (USA). Jaime Nivala thanks the Helmholtz Impulse and Networking Fund through Helmholtz Interdisciplinary Graduate School for Environmental Research (HIGRADE) and the Federal Ministry for Education and Research (BMBF) of Germany for funding and support. Joan García acknowledges the support from the Spanish Ministry of Education and Science; projects NEWWET (CTM2005-06457) and NEWWET2008 (CTM2008-06676). Joan García also appreciates very much the cooperation of undergraduate students, technicians and colleagues involved in these projects. The support of other private and public institutions is also greatly acknowledged.

## References

Álvarez, J.A., Ruiz, I., Soto, M., 2008. Anaerobic digesters as a pretreatment for constructed wetlands. *Ecol. Eng.* 33 (1), 54–67.

Asaeda, T., Nam, L.H., Heietz, P., Tanaka, N., Karunatatne, S., 2002. Seasonal fluctuations in live and dead biomass of *Phragmites australis* as described by a growth and decomposition model: implications of duration of aerobic conditions for litter mineralization and sedimentation. *Aquat. Bot.* 73 (3), 223–239.

Austin, D.C., 2005. Patent: Tidal vertical flow wastewater treatment system and method. United States: US 6,896,805 B2.

Baird, A.J., Surridge, B.W., Money, R.P., 2004. An assessment of the piezometer method for measuring the hydraulic conductivity of a *Cladium mariscus*-*Phragmites australis* root mat in a Norfolk (UK) fen. *Hydrol. Process.* 18, 275–291.

Barros, P., Ruiz, I., Soto, M., 2008. Performance of an anaerobic digester-constructed wetland system for a small community. *Ecol. Eng.* 33 (2), 142–149.

Batchelor, A., Loots, P., 1997. A critical evaluation of a pilot scale subsurface flow wetland: ten years after commissioning. *Water Sci. Technol.* 35 (5), 337–343.

Baveye, P., Vandevivere, P., Hoyle, B.L., DeLeo, P.C., de Lozada, D.S., 1998. Environmental impact and mechanisms of the biological clogging of saturated soils and aquifer materials. *Crit. Rev. Environ. Sci. Tech.* 28 (2), 123–191.

Bavor, H.J., Schulz, T.J., 1993. Sustainable suspended solids and nutrient removal in large-scale, solid-matrix, constructed wetland systems. In: Moshiri, G.A. (Ed.), *Constructed Wetlands for Water Quality Improvement*. Lewis Publishers, Boca Raton, FL, pp. 646–656.

Behrends, L.L., Houke, L., Bailey, E., Jansen, P., Brown, D., 2001. Reciprocating constructed wetlands for treating industrial, municipal, and agricultural wastewater. *Water Sci. Technol.* 44 (11–12), 399–405.

Blazejewski, R., Murat-Blazejewska, S., 1997. Soil clogging phenomena in constructed wetlands with subsurface flow. *Water Sci. Technol.* 35 (5), 183–188.

Boutin, C., Liénard, A., Esser, D., 1997. Development of a new generation of reedbed filters in France: first results. *Water Sci. Technol.* 35 (5), 315–322.

Bowmer, K.H., 1987. Nutrient removal from effluents by artificial wetland: influence of rhizosphere aeration and preferential flow studied using bromide and dye tracers. *Water Res.* 21 (5), 591–599.

Breen, P.F., Chick, A.J., 1995. Root zone dynamics in constructed wetlands receiving wastewater: a comparison of vertical and horizontal flow systems. *Water Sci. Technol.* 32 (3), 281–290.

Brix, H., 1994a. Functions of macrophytes in constructed wetlands. *Water Sci. Technol.* 29 (4), 71–78.

Brix, H., 1994b. Use of constructed wetlands in water pollution control: historical development, present status, and future perspectives. *Water Sci. Technol.* 30 (8), 209–223.

Brix, H., Arias, C.A., 2005. Danish guidelines for small-scale constructed wetland system for onsite treatment of domestic sewage. *Water Sci. Technol.* 51 (9), 1–9.

Brix, H., Johansen, N.H., 2004. Danish guidelines for vertical flow wetland systems (in Danish: Retningslinier for etablering af beplantede filteranlæg op til 30 PE). Århus, Denmark.

Brix, H., Schierup, H., 1989. Sewage treatment in constructed reed beds: Danish experiences. *Water Sci. Technol.* 21, 1665–1668.

Campbell, C.S., Ogden, M.H., 1999. *Constructed Wetlands in the Sustainable Landscape*. John Wiley & Sons, New York, New York.

Caselles-Osorio, A., García, J., 2006. Performance of experimental horizontal subsurface flow constructed wetlands fed with dissolved or particulate organic matter. *Water Res.* 40 (19), 3603–3611.

Caselles-Osorio, A., García, J., 2007. Effect of physico-chemical pretreatment on the removal efficiency of horizontal subsurface-flow constructed wetlands. *Environ. Pollut.* 146, 55–63.

Caselles-Osorio, A., Puigagut, J., Segu, E., Vaello, N., Granes, F., García, D., García, J., 2007. Solid accumulation in six full-scale subsurface flow constructed wetlands. *Water Res.* 41 (6), 1388–1389.

Chazarenc, F., Maltais-Landry, G., Troesch, S., Comeau, Y., Brisson, J., 2007. Effect of loading rate on performance of constructed wetlands treating an anaerobic supernatant. *Water Sci. Technol.* 56 (3), 23–29.

Chazarenc, F., Meriin, G., 2005. Influence of surface layer on hydrology and biology of gravel bed vertical flow constructed wetlands. *Water Sci. Technol.* 51 (9), 91–97.

Chimney, M.J., Pietro, K.C., 2006. Decomposition of macrophyte litter in a subtropical constructed wetland in south Florida (USA). *Ecol. Eng.* 27 (4), 301–321.

Christensen, J.B., Jensen, D.L., Grøn, C., Filip, Z., Christensen, T.H., 1998. Characterization of the dissolved organic carbon in landfill leachate-polluted groundwater. *Water Res.* 32 (1), 125–135.

Conley, L.M., Dick, R.L., Lion, L.W., 1991. An assessment of the root zone method of wastewater treatment. *J. Water Pollut. Control Fed.* 63 (3), 239–247.

Coombs, C., 1990. Reed bed treatment systems in Anglian water. In: Cooper, P.F., Findlater, B.C. (Eds.), *Constructed Wetlands in Water Pollution Control*. Pergamon Press, Oxford, United Kingdom, pp. 223–234.

Cooper, D., 2007. The Constructed Wetland Association UK database of constructed wetland systems. *Water Sci. Technol.* 56 (3), 1–6.

Cooper, D., Griffin, P., Cooper, P.F., 2008. Factors affecting the longevity of sub-surface horizontal flow systems operating as tertiary treatment for sewage effluent. In: Vymazal, J. (Ed.), *Wastewater Treatment, Plant Dynamics and Management in Constructed and Natural Wetlands*. Backhuys Publishers, Leiden, The Netherlands, pp. 191–198.

Cooper, D.J., Griffin, P., Cooper, P.F., 2005. Factors affecting the longevity of sub-surface horizontal flow systems operating as tertiary treatment for sewage effluent. *Water Sci. Technol.* 51 (9), 127–135.

Cooper, P.F., 2005. The performance of vertical-flow constructed wetland systems with special reference to the significance of oxygen transfer and hydraulic loading rate. *Water Sci. Technol.* 51 (9), 81–90.

Cooper, P.F., Cooper, D., 2005. Evaluation of a tidal flow reed bed system for the treatment of domestic sewage: nitrification trials. In: Vymazal, J. (Ed.), *Natu-*

- ral and Constructed Wetlands: Nutrients, Metals and Management. Backhuys Publishers, Leiden, The Netherlands, pp. 222–232.
- Cooper, P.F., Green, B., 1995. Reed bed treatment systems for sewage treatment in the United Kingdom: the first 10 years experience. *Water Sci. Technol.* 32 (3), 317–322.
- Cooper, P.F., Job, J.D., Green, B., Shutes, R.B.E., 1996. Reed Beds and Constructed Wetlands for Wastewater Treatment. WRC Publications, Swindon, United Kingdom, 184 p. plus data diskette.
- Corzo, A., Pedescoll, A., Álvarez, E., García, J., 2008. Solids accumulation and drainable porosity in experimental subsurface flow constructed wetlands with different primary treatments and operating strategies. In: Billore, S.K., Dass, P., Vymazal, J. (Eds.), Proceedings of the 11th International Conference on Wetland Systems for Water Pollution Control, 1–7 November 2008. Vikram University and IWA, Indore, India, pp. 290–295.
- Crites, R., Tchobanoglous, G., 1998. Small and Decentralized Wastewater Management Systems. McGraw-Hill, New York, New York.
- CWA Database, 2006. Constructed Wetlands Interactive Database. Compiled by G.D. Job and P.F. Cooper. Version 9.02.
- deBeer, D., Stoodley, P., 1995. Relation between the structure of an aerobic biofilm and transport phenomena. *Water Sci. Technol.* 32 (8), 11–18.
- Dornelas, F.L., Machado, M.B., von Sperling, M., 2008. Performance evaluation of planted and unplanted subsurface-flow constructed wetlands for the post-treatment of UASB reactor effluents. In: Billore, S.K., Dass, P., Vymazal, J. (Eds.), Proceedings of the 11th International Conference on Wetland Systems for Water Pollution Control, 1–7 November 2008. Vikram University and IWA, Indore, India, pp. 400–407.
- Dotro, G., Larsen, D., Palazolo, P., 2010. Preliminary evaluation of biological and physical-chemical chromium removal mechanisms in gravel media used in constructed wetlands. *Water Air Soil Pollut.*, doi 10.1007/s11270-010-0495-9.
- Drury, W.J., Mainzhausen, K., 2000. Hydraulic characteristics of subsurface flow wetlands. In: Proceedings of the 2000 Billings Land Reclamation Symposium, Billings, MT, USA.
- Dupin, H.J., Kitandis, P.K., McCarty, P.L., 2001. Simulations of two-dimensional modeling of biomass aggregate growth in network models. *Water Resour. Res.* 37 (12), 2981–2994.
- Dupin, H.J., McCarty, P.L., 1999. Mesoscale and microscale observations of biological growth in a silicon pore imaging element. *Environ. Sci. Technol.* 33 (8), 1230–1236.
- EC/EWPCA Emergent Hydrophyte Treatment Systems Expert Contact Group, Water Res Centre, 1990. In: P.F. C. (Ed.), European Design and Operations Guidelines for Reed Bed Treatment Systems. WRC, Swindon, United Kingdom.
- Fisher, P.J., 1990. Hydraulic characteristics of constructed wetlands at Richmond, New South Wales, Australia. In: Cooper, P.F., Findlater, B.C. (Eds.), Constructed Wetlands in Water Pollution Control. Pergamon Press, Oxford, United Kingdom, pp. 21–32.
- Fleming, I.R., Rowe, R.K., Cullimore, D.R., 1999. Field observations of clogging in a landfill leachate collection system. *Can. Geotech. J.* 36 (4), 685–707.
- García, J., Caselles-Osorio, A., Story, A., DePauw, N., Vanrolleghem, P.A., 2007. Impact of prior physico-chemical treatment on the clogging process of subsurface-flow constructed wetlands: model-based evaluation. *Water Air Soil Pollut.* 185, 101–109.
- García, J., Corzo, A., 2008. Depuración con Humedales Construidos: Guía Práctica de Diseño, Construcción y Explotación de Sistemas de Humedales de Flujo Subsuperficial. Universitat Politècnica de Catalunya, Barcelona, Spain.
- García, J., Ojeda, E., Sales, E., Chico, F., Piriz, T., Aguirre, P., Mujeriego, R., 2003. Spatial variations of temperature, redox potential, and contaminants in horizontal flow reed beds. *Ecol. Eng.* 21 (2–3), 129–142.
- Gersberg, R.M., Elkins, B.V., Goldman, C.R., 1984. Use of artificial wetlands to remove nitrogen from wastewater. *J. Water Pollut. Control Fed.* 56, 152–156.
- GFA (Gesellschaft zur Förderung der Abwassertechnik e.V.), 1998. Design Manual ATV-A 262: Principles of Dimensioning, Construction and Operation of Constructed Wetlands for Community Wastewater up to 1000 Person Equivalents (In German: Arbeitsblatt ATV-A 262: Grundsätze für Bemessung, Bau und Betrieb von Pflanzenbeetwn für kommunales Abwasser bei Ausbaugrößen bis 1000 Einwohnerwerte). St. Augustin, Germany.
- Green, M., Shaul, N., Beliaevski, M., Sabbah, I., Ghattas, B., Tarre, S., 2006. Minimizing land requirement and evaporation in small wastewater treatment systems. *Ecol. Eng.* 26 (3), 266–271.
- Green, M.B., Upton, J., 1995. Constructed reed beds: appropriate technology for small communities. *Water Sci. Technol.* 32 (3), 339–348.
- Griffin, P., Pamplin, C., 1998. The advantages of a constructed reed bed based strategy for small sewage treatment works. *Water Sci. Technol.* 38 (3), 143–150.
- Griffin, P., Wilson, L., Cooper, D., 2008. Changes in the use, operation and design of sub-surface flow constructed wetlands in a major UK water utility. In: Billore, S.K., Dass, P., Vymazal, J. (Eds.), Proceedings of the 11th International Conference on Wetland Systems for Water Pollution Control, 1–7 November 2008. Vikram University and IWA, Indore, India, pp. 419–426.
- Haberl, R., Perfler, J., 1990. Seven years of research work and experience with wastewater treatment by a reed bed system. In: Cooper, P.F., Findlater, B.C. (Eds.), Constructed Wetlands in Water Pollution Control. Pergamon Press, Oxford, United Kingdom, pp. 205–214.
- Hermansson, M., 1999. The DLVO theory in microbial adhesion. *Biointerfaces* 14 (1–4), 105–119.
- Hubbe, M.A., Chen, H., Heitmann, J.A., 2009. Permeability reduction phenomena in packed beds, fiber mats, and wet webs of paper exposed to flow of liquids and suspensions: a review. *BioResources* 4 (1), 405–451.
- Hyánková, E., Křiška-Dunajský, M., Rozkošný, M., Sálek, J., 2006. The knowledge based on the research of the filtration properties of the filter media and on the determination of clogging causes. In: Proceedings of the 10th International Conference on Wetland Systems for Water Pollution Control, 23–29 September 2006. Ministério de Ambiente, do Ordenamento do Território e do Desenvolvimento Regional (MAOTDR) and IWA, Lisbon, Portugal, pp. 1331–1338.
- IWA Specialist Group on Use of Macrophytes in Water Pollution Control, 2000. Constructed Wetlands for Pollution Control: Processes, Performance, Design and Operation. London, United Kingdom: IWA Publishing.
- Iwema, A., Raby, D., Lesarve, J., Boutin, C., Dodane, P.-H., Liénard, A., Molle, P., Beck, C., Sadowski, G.A., Merlin, G., Dap, S., Ohresser, C., Poulet, J.-B., Reeb, G., Werckmann, M., Esser, D., 2005. Treatment of domestic waste water plant with filters macrophytes: technical recommendations for the design and realization. (In French: Épuration des eau usées domestiques par filtres plantes de macrophytes: Recommandations techniques pour la conception et la réalisation). Agences de l'Eau Rhône, Méditerranée et Course, et Rhin Meuse, France.
- Kadlec, R.H., Knight, R.L., 1996. Treatment Wetlands, first ed. CRC Press, Boca Raton, FL.
- Kadlec, R.H., Wallace, S.D., 2009. Treatment Wetlands, second ed. CRC Press, Boca Raton, FL.
- Kadlec, R.H., Watson, J.T., 1993. Hydraulics and solids accumulation in a gravel bed treatment wetland. In: Moshiri, G.A. (Ed.), Constructed Wetlands for Water Quality Improvement. Lewis Publishers, Boca Raton, FL, pp. 227–235.
- Kayser, K., Kunst, S., 2005. Processes in vertical-flow reed beds: nitrification, oxygen transfer, and soil clogging. *Water Sci. Technol.* 51 (9), 177–184.
- Khatiwada, N.R., Polprasert, C., 1999. Assessment of effective specific surface area for free water surface constructed wetlands. *Water Sci. Technol.* 40 (3), 83–89.
- Kickuth, R.W., Könnemann, N., 1988. Patent: Method for the purification of sewage waters. United States: US 4,793,929.
- King, A.C., Mitchell, C.A., Howes, T., 1997. Hydraulic tracer studies in a pilot scale subsurface flow constructed wetland. *Water Sci. Technol.* 35 (5), 189–196.
- Kirschner, A., Riegl, B., Velimirov, B., 2001. Degradation of emergent and submerged macrophytes in an oxbow lake of an embanked backwater system: implications for the terrestrialization process. *Int. Rev. Hydrobiol.* 86 (4–5), 555–571.
- Knowles, P.R., Davies, P.A., 2009. A method for the in-situ determination of the hydraulic conductivity of gravels as used in constructed wetlands for wastewater treatment. *Desalination Water Treat.* 1 (5), 257–266.
- Knowles, P.R., Griffin, P., Davies, P.A., 2010. Complementary methods to investigate the development of clogging within a horizontal subsurface flow tertiary treatment wetland. *Water Res.* 44 (1), 320–330.
- Lahav, O., Artzi, E., Tarre, S., Green, M., 2001. Ammonium removal using a novel unsaturated flow biological filter with passive aeration. *Water Res.* 35 (2), 397–404.
- Langergraber, G., 2008. Modeling of processes in subsurface flow constructed wetlands: a review. *Vadosa Zone J.* 7 (2), 830–842.
- Langergraber, G., Haberl, R., Laber, J., Pressl, A., 2003. Evaluation of substrate clogging processes in vertical flow constructed wetlands. *Water Sci. Technol.* 48 (5), 25–34.
- Levine, A.D., Tchobanoglous, G., Asano, T., 1991. Size distributions of particulate contaminants in wastewater and their impact on treatability. *Water Res.* 25 (8), 911–922.
- Liénard, A., Boutin, C., Esser, D., 1990. Domestic wastewater treatment with emergent hydrophyte beds in France. In: Cooper, P.F., Findlater, B.C. (Eds.), Constructed Wetlands in Water Pollution Control. Pergamon Press, Oxford, United Kingdom, pp. 183–192.
- Llorens, E., Puigagut, J., García, J., 2009. Distribution and biodegradability of sludge accumulated in a full-scale horizontal subsurface-flow constructed wetland. *Desal. Water Treat.* 4, 54–58.
- Logan, B.E., Jewett, D.G., Arnold, R.G., Bouwer, E.J., O'Melia, C.R., 1995. Clarification of clean-bed filtration models. *J. Environ. Eng.* 121 (12), 837–869.
- Madigan, M.T., Martinko, J.M., Brock, T.D., 2006. Brock Biology of Microorganisms, 11th ed. Pearson Prentice Hall, Upper Saddle River, New Jersey.
- Maloszewski, P., Wachniew, P., Czuprynski, P., 2006. Study of hydraulic parameters in heterogeneous gravel beds: constructed wetland in Nowa Słupia (Poland). *J. Hydrol.* 331 (3–4), 630–642.
- Mays, D.C., Hunt, R.J., 2005. Hydrodynamic aspects of particle clogging in porous media. *Environ. Sci. Technol.* 39 (2), 577–584.
- Molle, P., Liénard, A., Boutin, C., Merlin, G., Iwema, A., 2005. How to treat raw sewage with constructed wetlands: an overview of the French systems. *Water Sci. Technol.* 51 (9), 11–21.
- Molle, P., Liénard, A., Grasmick, A., Iwema, A., 2006. Effect of reeds and feeding operations on hydraulic behaviour of vertical flow constructed wetlands under hydraulic overloads. *Water Res.* 40 (3), 606–612.
- Muñoz, P., Drizo, A., Hession, W.C., 2006. Flow patterns of dairy wastewater constructed wetlands in a cold climate. *Water Res.* 40 (17), 3209–3218.
- Murphy, C., Cooper, D., 2010. The evolution of horizontal sub-surface flow reed bed design for tertiary treatment of sewage effluents in the UK. In: Vymazal, J. (Ed.), Water and Nutrient Management in Natural and Constructed Wetlands. Springer, Dordrecht, Netherlands.
- Netter, R., Bischofsberger, W., 1990. Hydraulic investigations on planted soil filters. In: Cooper, P.F., Findlater, B.C. (Eds.), Constructed Wetlands in Water Pollution Control. Pergamon Press, Oxford, United Kingdom, pp. 11–20.

- Nguyen, L., 2001. Accumulation of organic matter fractions in a grave-bed constructed wetland. *Water Sci. Technol.* 44 (11–12), 281–287.
- Nguyen, L.M., 2000. Organic matter composition, microbial biomass and microbial activity in gravel-bed constructed wetlands treating farm dairy wastewaters. *Ecol. Eng.* 16 (2), 199–221.
- Nivala, J.A., Hoos, M.B., Cross, C.S., Wallace, S.D., Parkin, G.F., 2007. Treatment of landfill leachate using an aerated, horizontal subsurface-flow constructed wetland. *Sci. Total Environ.* 380, 19–27.
- Nivala, J.A., Rousseau, D.P.L., 2009. Reversing clogging in subsurface-flow constructed wetlands by hydrogen peroxide treatment: two case studies. *Water Sci. Technol.* 59 (10), 2037–2046.
- ÖNORM B 2505, 2005. Subsurface flow constructed wetlands – Application, dimensioning, installation, and operation (Update to 1997 publication) (In German: Bepflanzte Bodenfilter (Pflanzenkläranlagen) – Anwendung, Bemessung, Bau und Betrieb). Österreichisches Normungsinstitut: Vienna, Austria.
- Parr, T.W., 1990. Factors affecting reed (*Phragmites australis*) growth in UK reed bed treatment systems. In: Cooper, P.F., Findlater, B.C. (Eds.), *Constructed Wetlands in Water Pollution Control*. Pergamon Press, Oxford, United Kingdom, pp. 67–76.
- Pauly, U., 1990. Performance data of a wastewater and sludge treatment plant derived from the Root Zone Method set against the background of detention times. In: Cooper, P.F., Findlater, B.C. (Eds.), *Constructed Wetlands in Water Pollution Control*. Pergamon Press, Oxford, United Kingdom, pp. 289–300.
- Pedescoll, A., Uggetti, E., Llorens, E., Granés, F., García, D., García, J., 2009. Practical method based on hydraulic conductivity used to assess clogging in subsurface flow constructed wetlands. *Ecol. Eng.* 35 (8), 1216–1224.
- Platzer, C., Mauch, K., 1997. Soil clogging in vertical flow reed beds: mechanisms, parameters, consequences and.. solutions? *Water Sci. Technol.* 35 (5), 175–182.
- Puigagut, J., Caselles-Osorio, A., Vaello, N., García, J., 2008. In: Vymazal, J. (Ed.), *Fractionation, Biodegradability, and Particle-size Distribution of Organic Matter in Horizontal Subsurface-flow Constructed Wetlands*. Springer, Dordrecht, Netherlands, pp. 289–297.
- Ragusa, S.R., McNeven, D., Qasem, S., Mitchell, C., 2004. Indicators of biofilm development and activity in constructed wetlands microcosms. *Water Res.* 38, 2865–2873.
- Reed, S.C., Crites, R., Middlebrooks, E.J., 1995. *Natural Systems for Waste Management and Treatment*. McGraw-Hill, New York.
- Rittman, B.E., McCarty, P.L., 1980. Evaluation of steady-state biofilm kinetics. *Biotechnol. Bioeng.* 22 (11), 2359–2373.
- Rousseau, D.P.L., Horton, D., Vanrolleghem, P.A., De Pauw, N., 2005. Impact of operational maintenance on the asset life of storm reed beds. *Water Sci. Technol.* 51 (9), 243–250.
- Rybaczek, J.M., Day, J.W., Conner, W.H., 2002. The impact of wastewater effluent on accretion and decomposition in a subsiding forested wetland. *Wetlands* 22 (1), 18–32.
- Sanford, W.E., Steenhuis, T.S., Parlange, J.Y., Surface, J.M., Peverly, J.H., 1995. Hydraulic conductivity of gravel and sand as substrates in rock-reed filters. *Ecol. Eng.* 4 (4), 321–326.
- Seidel, K., 1973. Patent: System for purification of polluted water. United States: US 3,770,623.
- Sheoran, A., Sheoran, V., 2006. Heavy metal removal mechanism of acid mine drainage in wetlands: a critical review. *Miner. Eng.* 19, 105–116.
- Spangler, F.L., Sloey, W.E., Fetter, C.W.J., 1976. *Wastewater Treatment by Natural and Artificial Marshes*. US EPA Office of Water, Ada, Oklahoma.
- Steiner, G.R., Freeman, R.J., 1989. In: Hammer, D.A. (Ed.), *Configuration and Substrate Design Considerations for Constructed Wetlands for Wastewater Treatment*. Lewis Publishers, Chelsea, MI, pp. 363–377.
- Suliman, F., French, H.K., Haugen, L.E., Søvik, A.K., 2006. Change in flow and transport patterns in horizontal subsurface flow constructed wetlands as a result of biological growth. *Ecol. Eng.* 27 (2), 124–133.
- Sun, G., Zhao, Y., Allen, S., 2007. An alternative arrangement of gravel media in tidal flow reed beds treating pig farm wastewater. *Water Air Soil Pollut.* 182 (1), 13–19.
- Swift, D.L., Friedlander, S.K., 1964. The coagulation of hydrosols by Brownian motion and laminar shear flow. *J. Coll. Sci. Imp. U. Tok.* 19 (7), 621–647.
- Tanner, C.C., 1994. Growth and nutrition of *Schoenoplectus validus* in agricultural wastewaters. *Aquat. Bot.* 47 (2), 131–153.
- Tanner, C.C., Sukias, J.P., 1995. Accumulation of organic solids in gravel bed constructed wetlands. *Water Sci. Technol.* 32 (3), 229–239.
- Tanner, C.C., Sukias, J.P.S., Upsdell, M.P., 1998. Organic matter accumulation and maturation of gravel bed constructed wetlands treating dairy farm wastewaters. *Water Res.* 32 (10), 3046–3054.
- Taylor, S.W., Milly, P.C.D., Jaffe, P.R., 1990. Biofilm growth and the related changes in the physical properties of a porous medium: 2. Permeability. *Water Resour. Res.* 26 (9), 2161–2169.
- Tchobanoglous, G., 1993. Constructed wetlands and aquatic plant systems: research, design, operational and monitoring issues. In: Moshiri, G.A. (Ed.), *Constructed Wetlands for Water Quality Improvement*. Lewis Publishers, Boca Raton, FL, pp. 23–34.
- Tchobanoglous, G., 2003. Preliminary treatment in constructed wetlands. In: Proceedings of the 1st International Seminar on the Use of Aquatic Macrophytes for Wastewater Treatment in Constructed Wetlands, 8–10 May 2003. Instituto da Conservação da Natureza and Instituto da Água: Lisbon, Portugal, pp. 13–33.
- Thullner, M., 2010. Comparison of bioclogging effects in saturated porous media within one- and two-dimensional flow systems. *Ecol. Eng.* 36 (2), 176–196.
- Tietz, A., Kirschner, A., Langergraber, G., Sleytr, K., Haberl, R., 2007. Characterisation of microbial biocenosis in vertical subsurface flow constructed wetlands. *Sci. Total Environ.* 380, 163–172.
- Tufenkji, N., 2007. Modelling microbial transport in porous media: traditional approaches and recent developments. *Adv. Water Resour.* 30 (6–7), 1455–1469.
- TVA, 1993. General design, construction, and operation guidelines: constructed wetlands wastewater treatment systems for small users including individual residences, 2nd ed. In: Steiner, G.R., Watson, J.T. (Eds.), *Tennessee Valley Authority (TVA) Resource Group Water Management: Chattanooga, Tennessee*.
- US EPA, 1988. Design Manual: Constructed Wetlands and Aquatic Plant Systems For Municipal Wastewater Treatment. US EPA Office of Water, Cincinnati, Ohio.
- US EPA, 1993. Subsurface Flow Constructed Wetlands For Wastewater Treatment: A Technology Assessment. US EPA Office of Water.
- US EPA, 2000. Constructed Wetlands Treatment of Municipal Wastewaters. US EPA Office of Research and Development, Washington, DC.
- US EPA, 2002. Onsite Wastewater Treatment Systems Manual. US EPA Office of Research and Development, Washington, DC.
- Vandevivere, P., Baveye, P., 1992a. Effect of bacterial extracellular polymers on the saturated hydraulic conductivity of sand columns. *Appl. Environ. Microb.* 58 (5), 1690–1698.
- Vandevivere, P., Baveye, P., 1992b. Relationship between transport of bacteria and their clogging efficiency in sand columns. *Appl. Environ. Microb.* 58 (8), 2523–2530.
- Vandevivere, P., Baveye, P., 1992c. Saturated hydraulic conductivity reduction caused by aerobic bacteria in sand columns. *Soil Sci. Soc. Am. J.* 56 (1), 1–13.
- Vymazal, J., 1996. The use of subsurface flow constructed wetlands for wastewater treatment in the Czech Republic. *Ecol. Eng.* 7, 1–14.
- Vymazal, J., Brix, H., Cooper, P.F., Green, M.B., Haberl, R., 1998. *Constructed Wetlands for Wastewater Treatment in Europe*. Backhuys Publishers, Leiden, The Netherlands.
- Vymazal, J., Kröpfelová, L., 2008. *Wastewater Treatment in Constructed Wetlands with Horizontal Sub-Surface Flow*. Springer.
- Wallace, S.D., Knight, R.L., 2006. Small-scale constructed wetland treatment systems: feasibility, design criteria, and O&M requirements. *Water Environment Research Foundation (WERF)*, Alexandria, Virginia.
- Waters, M., Pilgrim, D.H., Schulz, T.J., Pilgrim, I.D., 1993. Variability of hydraulic response of constructed wetlands. In: Proceedings of the National Conference on Hydraulic Engineering, 25–30 July 1993 (San Francisco, California). American Society of Civil Engineers (ASCE), pp. 406–411.
- Watson, J.T., Choate, K.D., Steiner, G.R., 1990. Performance of constructed wetland treatment systems at Benton, Hardin, and Pembroke, Kentucky, during the early vegetation establishment phase. In: Cooper, P.F., Findlater, B.C. (Eds.), *Constructed Wetlands in Water Pollution Control*. Pergamon Press, Oxford, United Kingdom, pp. 171–182.
- Watson, J.T., Hobson, J.A., 1989. In: Hammer, D.A. (Ed.), *Hydraulic Design Considerations and Control Structure For Constructed Wetlands For Wastewater Treatment*. Lewis Publishers, Chelsea, Michigan, pp. 379–391.
- WERF Database, 2006. Small-Scale Constructed Wetland Treatment Systems Database (Project 01-CTS-5; Final Report by S.D. Wallace and R.L. Knight, 2006). Compiled by J.A. Nivala and R.A. Clarke.
- Winter, K.-J., Goetz, D., 2003. The impact of sewage composition on the soil clogging phenomena of vertical flow constructed wetlands. *Water Sci. Technol.* 48 (5), 9–14.
- Yao, K.-M., Habibian, M.T., O'Melia, C.R., 1971. Water and waste water filtration: concepts and applications. *Environ. Sci. Technol.* 5 (11), 1105–1112.
- Zachritz, W.H.L., Fuller, J.W., 1993. Performance of An Artificial Wetlands Filter Treating Facultative Lagoon Effluent at Carville, vol. 65, no. 1. *Water Environment Research*, Louisiana, pp. 46–52.
- Zamani, A., Maini, B., 2009. Flow of dispersed particles through porous media: deep bed filtration. *J. Petrol. Sci. Eng.* 69 (1–2), 71–88.
- Zhao, Y.Q., Sun, G., Allen, S.J., 2004. Anti-sized reed bed system for animal wastewater treatment: a comparative study. *Water Res.* 38, 2907–2917.
- Zhu, T., Jessen, P.D., Møhlum, T., Krogstad, T., 1997. Phosphorus sorption and chemical characteristics of lightweight aggregates (LWA)—potential filter media in treatment wetlands. *Water Sci. Technol.* 35 (5), 103–108.
- Zoeller, K.E., Byers, M.E., 1999. Patent: Wastewater treatment system. United States: US 5,897,777.